OntoCoDec: Type-Constrained Information Extraction via Compiled Schema Constraints

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Abstract

We present OntoCoDec, a framework that translates Information Extraction (IE) ontologies into composable token-level decoding constraints. A decidability analysis yields a two-layer architecture: a *grammar layer* (context-free) compiles ontology constraints into grammars ensuring JSON validity and vocabulary membership; a *semantic layer* (prefix-decidable with state) interprets type constraints at runtime via stateful token masking over a dynamic entity registry, with grammar-first fallback when constraints conflict. Composed with LoRA fine-tuning on three 8B-class models across two RE datasets (up to 96 relation types), OntoCoDec near-eliminates domain/range violations (DR $< 0.1\%$) at R-F1 41–47%. This enforcement introduces cross-field violations (CF) as a side effect—up to 1.18% under greedy decoding on well-calibrated models—because per-triple enforcement cannot guarantee global entity-type consistency; beam search (k=4) eliminates all violation types at 1.54× latency. Zero-shot evaluation reveals DR reduction of 76–99% on SciERC and 42–90% on DocRED, with a scale-dependent enabling dependency: neither layer achieves full DR reduction alone at 8B, while 70B semantic-only reaches 84%. 70B-class validation on SciERC (full six-way ablation) with preliminary DocRED support, and event extraction on WikiEvents, confirm generalizability.

1 Introduction

Large language models are increasingly used for information extraction (Li et al., 2023; Sainz et al., 2024; Wei et al., 2023), but when generating relation triples (h, r, t), neither entity necessarily carries a type consistent with the ontology's domain/range constraints for r. We call such outputs *domain/range violations*: structurally valid triples that are ontologically invalid. Violations persist across the capability spectrum: 15.4% of triples on SciERC (Luan et al., 2018) and 18.5% on DocRED (Yao et al., 2019) under zero-shot decoding, and 0.5–2% even after LoRA fine-tuning; rates correlate with schema restrictiveness (few valid type pairs per relation) rather than ontology or model scale. At competitive quality (R-F1 41–47%), even 1–2% residual violations compound across thousands of documents into systematic knowledge graph corruption, making reliable constraint enforcement necessary.

Existing approaches leave a fundamental gap. Schema-informed prompting reduces violations by 62% on average but is per-relation unreliable (violations for EVALUATE-FOR actually *increase* by 5.8 pp). Grammar-constrained decoding (Dong et al., 2025; Willard and Louf, 2023; Beurer-Kellner et al., 2023) ensures syntactic well-formedness but cannot encode cross-field semantic constraints. Post-hoc filtering discards up to 48% of predictions including true positives (§5.3). Training-based approaches (Sainz et al., 2024; Lu et al., 2022) internalize schema knowledge but provide only soft guidance without runtime enforcement.

We propose **OntoCoDec**, a framework that translates IE ontologies into composable layers of decoding-time constraints. A decidability analysis of IE type constraints (§3) yields a two-layer architecture: the *grammar layer* compiles ontology constraints into context-free grammars (JSON structure, vocabulary membership) via XGrammar; the *semantic layer* interprets type constraints at runtime via dynamic logit masking over an entity registry, with a scale-dependent enabling dependency: at 8B, the semantic layer achieves only 24% independent DR reduction (near-complete when combined), while at 70B it reaches 84% independently. OntoCoDec compiles constraints automatically from IE ontologies, requires no constraint-specific training, and is applicable to autoregressive LMs with logits access; we validate on three English

datasets (up to 96 relation types), three 8B-class models across three tokenizer families, with 70B-class validation (§5.4). LoRA + greedy S1 achieves near-complete DR elimination: DR $\leq 0.09\%$ at R-F1 41–47% (± 0.8 pp, $+10\%$ latency); beam (k=4) eliminates all violation types on both RE datasets under fine-tuning (1.54× latency). DR enforcement converts ontologically invalid violations to individually valid but type-inconsistent ones (CF $\leq 1.4\%$ fine-tuned); zero-shot DR drops 42–99% across models and datasets, with total violations decreasing $\sim$31% (§5.3).

We make a systematic engineering contribution with theoretical grounding:

1. A two-layer constrained decoding framework: we formalize a well-known intuition that cross-field type constraints require runtime state beyond context-free expressiveness (Proposition 1), yielding pair-based sequential enforcement at each level (§3). Constraints compile automatically from IE ontologies, validated on schemas of varying restrictiveness with up to 96 relation types.

2. A comprehensive evaluation across 3 IE datasets, 3 models (8B-class), and 2 tasks, with 70B validation (full SciERC ablation, preliminary DocRED): DR $\leq 0.09\%$ at R-F1 41–47%; zero-shot ablation reveals an enabling dependency (neither layer alone exceeds 24% DR reduction at 8B; 84% at 70B; §5.4); DR enforcement introduces CF as a correctable side effect (§5.1); beam eliminates all violations; WikiEvents validates event extraction applicability (§5.4).

2 Background

IE Ontology Structure. An IE ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$ defines entity types $\mathcal{E}$, relation types $\mathcal{R}$, and a mapping $\sigma : \mathcal{R} \to \mathcal{P}(\mathcal{E} \times \mathcal{E})$, where $\mathcal{P}(\cdot)$ denotes the power set, specifying valid (head type, tail type) pairs per relation. A triple (h, r, t) is *type-valid* if $(\text{type}(h), \text{type}(t)) \in \sigma(r)$. We quantify constraint restrictiveness via *constraint tightness* $\kappa(r)$ (§4): tighter constraints leave fewer valid type pairs per relation. We define ontology constraints broadly to encompass both formal ontology specifications (e.g., the KAIROS event ontology with explicit role-entity type mappings) and empirically derived type-pair constraints extracted from annotated training data. The latter applies when datasets lack a formal ontology but provide typed annotations (SciERC, DocRED); OntoCoDec compiles both sources into the same constraint representation without modification. Empirical constraints achieve 99%+ instance-level coverage on held-out data (SciERC 99.0%, DocRED 99.4%); uncovered type pairs trigger grammar-first fallback (§4), ensuring no valid triples are blocked.

Grammar-Constrained Decoding. Grammar-constrained decoding restricts each token y_t to a mask $M_g \in \{0, 1\}^{|V|}$ derived from a target grammar. XGrammar (Dong et al., 2025) compiles JSON Schemas into pushdown automata for efficient enforcement of structural constraints; Outlines (Willard and Louf, 2023) and SGLang (Zheng et al., 2024) offer similar capabilities. However, these methods are limited to context-free constraints and cannot enforce cross-field semantic rules (e.g., "if r = FEATURE-OF, then $\text{type}(h) \in \{\text{METHOD}, \text{OTHERSCIENTIFICTERM}\}$"), which depend on previously generated field values.

Problem Statement. Given $\mathcal{O}$ and an autoregressive LM $\mathcal{M}$, we seek constraints $\mathcal{C}$ ensuring: (a) **structural validity**, i.e., well-formed JSON with $\mathcal{E}/\mathcal{R}$ membership; and (b) **type validity**, i.e., $(\text{type}(h), \text{type}(t)) \in \sigma(r)$ for every triple. Condition (a) is context-free; (b) is context-sensitive but *prefix-decidable*: at the point where a type token is generated, all information needed (relation type, entity registry) is available in the decoded prefix. This requires that each entity type name maps to a distinguishable token sequence (§3.3).

3 Method

We present OntoCoDec, a framework that translates IE ontologies into token-level constraints enforced during autoregressive generation. The design is guided by a decidability analysis of IE type constraints.

Constraint decidability taxonomy. We classify IE type constraints into three levels: (1) *context-free*: vocabulary membership and JSON structure; (2) *prefix-decidable with state*: domain/range type validity, determined at each type-bearing position given the decoded relation type and entity registry; (3) *non-local*: cross-entity type consistency, which depends on the full output. Unlike code generation where types are statically declared, IE constraints fall into level 2: entity types are inferred

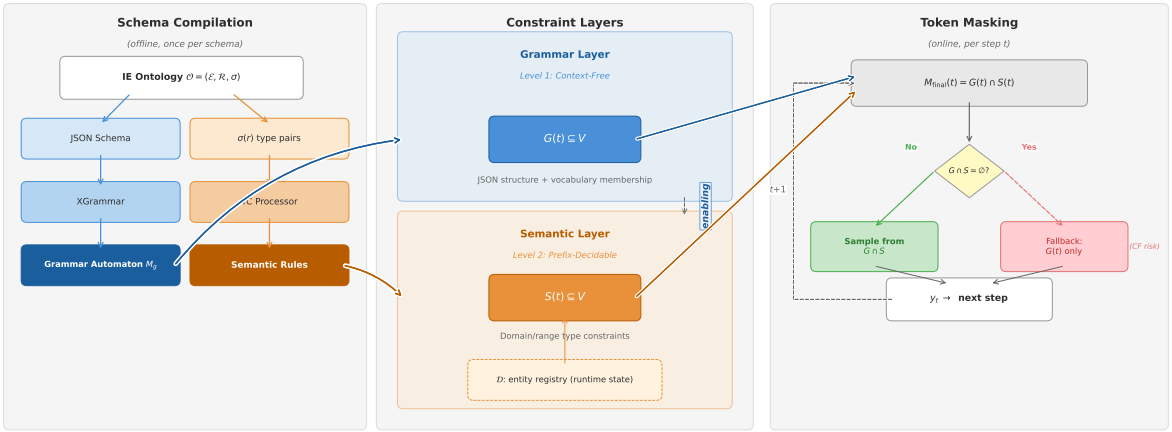


Figure 1: Overview of OntoCoDec. An IE ontology is compiled into two constraint layers: a grammar layer (context-free) and a semantic layer (prefix-decidable with state). At each decoding step, the final mask $M_{\text{final}}(t) = G(t) \cap S(t)$ ensures both structural and type validity; when $S(t) \cap G(t) = \emptyset$, grammar-first fallback preserves JSON integrity.

from text, the registry is built dynamically, and constraint targets depend on the co-occurring relation. OntoCoDec enforces levels 1–2 (grammar and semantic layers, respectively); level 3 is addressed by beam search or post-hoc resolution (Appendix M).

Existing grammar-constrained decoding frameworks (e.g., XGrammar) enforce level 1 constraints—structural validity and vocabulary membership—but IE domain/range rules require cross-position entity-type state that context-free grammars cannot express. We formalize this gap:

Proposition 1 (Level 2 $\not\subseteq$ Level 1)**.** *Domain/range type constraints for IE with unbounded entity sets are not context-free.*

Proof sketch. Consider $\mathcal{E} = \{\tau_A, \tau_B\}$ and a relation r with $\text{dom}(r) = \{\tau_A\}$. An output contains n entities with independently chosen types, followed by a relation referencing head e_k for arbitrary $k \leq n$. A PDA's LIFO stack cannot retrieve position $k < n$ without destroying entries above it; since both n and k are independently unbounded, no PDA can enforce this constraint. OntoCoDec's semantic layer addresses this with a runtime entity registry $\mathcal{D}$ (a finite map enabling random-access type lookups; Appendix O). □

Remark (finite entity sets)**.** When the entity set is finite (the practical case), the valid language is technically regular, but the corresponding DFA requires $O(|\mathcal{E}|^{|\text{entities}|})$ states—exceeding 10^{17} for DocRED (7 entity types, 20+ entities per document)—making direct DFA construction intractable. The distinction is between theoretical impossibility (unbounded) and engineering intractability (finite but exponential); OntoCoDec's registry-based approach avoids both.

Empirical Observation (scale-dependent enabling dependency)**.** The grammar layer alone cannot enforce domain/range constraints (Proposition 1), while the semantic layer alone achieves only 24% DR reduction at 8B without grammar-enforced vocabulary membership (§5.1, Table 1). The strength of this dependency is scale-dependent: at 70B, the semantic layer independently achieves 84% DR reduction (§5.4), whereas 8B models show only 24% independent reduction.

We describe the problem formulation (§3.1), each layer, and two decoding strategies (§3.4); formal analysis of constraint composition is in Section 4.

3.1 Problem Formulation

IE as structured generation. Given an input text x, a language model generates a structured JSON output y:

$$y = \{\texttt{entities} : [\{n_i, \tau_i\}],\ \texttt{relations} : [\{h_j, t_j, r_j, \tau_j^h, \tau_j^t\}]\} \quad (1)$$

where each entity has a name n_i and type $\tau_i \in \mathcal{E}$, and each relation specifies head h_j, tail t_j, relation type $r_j \in \mathcal{R}$, and head/tail entity types τ_j^h, τ_j^t. The output JSON Schema (§3.2) enforces a fixed field ordering (head → tail → relation → head_type → tail_type) via deterministic grammar compilation, ensuring the relation type r_j is known before entity type fields and enabling sequential domain/range lookups (§3.3). The grammar structurally enforces

entity-before-relation ordering (100% preservation across all configurations), ensuring the entity registry (§3.3) is populated before relation-level type constraints are evaluated; unregistered entities trigger domain-only filtering.

Ontology compliance. Using the ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$ defined in §2, we write $\mathrm{dom}(r) = \{\tau^h : \exists \tau^t,\ (\tau^h, \tau^t) \in \sigma(r)\}$ for domain-valid head types and $\mathrm{range}(r, \tau^h) = \{\tau^t : (\tau^h, \tau^t) \in \sigma(r)\}$ for the conditional range given head type τ^h. Our enforcement target is *pair-based* type validity: $(\tau^h, \tau^t) \in \sigma(r)$ for every triple. The decoder enforces this sequentially: $\tau^h \in \mathrm{dom}(r)$ at head position, then $\tau^t \in \mathrm{range}(r, \tau^h)$ at tail position, achieving best-effort pair-based enforcement (pair validity ensured when no fallback is triggered; domain-only filtering otherwise) strictly stronger than independent domain/range checks (which over-permit 33–37% spurious pairs on SciERC/DocRED; §4).

3.2 Grammar Layer

A JSON Schema encoding Eq. (1) with `enum` constraints ($\mathcal{E}$, $\mathcal{R}$) is compiled into a pushdown automaton via XGrammar (Dong et al., 2025). At step t, grammar-valid tokens $G(t) \subseteq V$ pass; all others receive logits $-\infty$, enforcing structural validity, vocabulary membership ($r_j \in \mathcal{R}$, $\tau_i \in \mathcal{E}$), but not domain/range constraints (which depend on entity types within each triple).

3.3 Semantic Layer

The semantic layer enforces domain/range constraints through a stateful logits processor maintaining an entity registry during generation. *Running example*: for USED-FOR ($\mathrm{dom} = \{\text{Method}\}$, $\mathrm{range} = \{\text{Task}\}$), entity "ResNet" (Method) passes the head-type check while Material is blocked; "image classification" (Task) passes the tail-type check.

Entity registry. The semantic layer builds a registry $\mathcal{D}$: name $\rightarrow$ type incrementally during entity generation: each completed entry $\{n_i, \tau_i\}$ adds (n_i, τ_i) to $\mathcal{D}$, which is consulted during relation generation for type lookups. At each step t, the processor pattern-matches the decoded prefix to identify the current field position; semantic masks activate only at type-bearing positions (HEADTYPE, TAILTYPE).

Dynamic masking. At type-bearing positions, the processor computes a semantic mask $S(t) \subseteq V$. At a HEADTYPE position, if $h_j \in \mathcal{D}$ and $\mathcal{D}[h_j] \in \mathrm{dom}(r_j)$, the mask allows only the registered type; otherwise it falls back to all domain-valid types $\mathrm{dom}(r_j)$. At a TAILTYPE position, the mask restricts to $\mathrm{range}(r_j, \tau_j^h)$ conditioned on the already-determined head type, further narrowing to the registered type when $t_j \in \mathcal{D}$. This sequential check achieves best-effort pair-based enforcement (§3.1), strictly tighter than independent domain/range checking; "best-effort" reflects the grammar-first fallback (§4), which prioritizes JSON integrity over type validity when constraints conflict.

Tokenizer-agnostic type masking. We use *continuation masking* to handle context-dependent tokenizations and prefix-shared types (e.g., SEN/SID/SIDE in WikiEvents): a compile-time prefix map identifies valid continuations for each type name, disambiguating over multiple steps across byte-level BPE (Qwen, Llama) and SentencePiece (Mistral).

Grammar-semantic interaction. The final mask is $G(t) \cap S(t)$; when empty, grammar-first fallback applies (§4), embodying the empirical enabling dependency: semantic constraints reduce DR by only 24% without grammar-enforced vocabulary.

3.4 Decoding Strategies

S1 (joint extraction): Single-pass decoding. Both layers are active throughout generation; the entity registry populates during entity decoding and is immediately available for relation-level type checking.

S2 (two-pass pipeline): Two-pass decoding. A first pass extracts entities; these compile an input-dependent grammar for the second pass. S1 achieves higher R-F1 at comparable DR (§5.1) and is the primary configuration.

4 Constraint Composition

At each decoding step t, let $G(t) \subseteq V$ and $S(t) \subseteq V$ denote the tokens permitted by the grammar and semantic layers, respectively. The final mask is their conjunction:

$$M_{\text{final}}(t) = M_{\text{grammar}}(t) \ \wedge \ M_{\text{semantic}}(t), \quad (2)$$

where $\wedge$ denotes element-wise AND over the vocabulary V; a token is allowed only if it satisfies *both* structural and type-level constraints. The conjunction is asymmetric: $S(t)$ is defined only over

$G(t)$. This reflects the empirical enabling dependency (§3): the semantic layer requires the relation type to belong to $\mathcal{R}$, a property enforced only by the grammar layer. This dependency is scale-sensitive: 70B semantic-only achieves 84% DR reduction (vs. 24% on 8B), though grammar remains necessary for RV elimination (§5.4).

Grammar-First Fallback. When $S(t) \cap G(t) = \emptyset$ (tight constraints leave no token that is both grammar-valid and semantically valid), naïve masking would force selection of a grammar-invalid token, breaking JSON structure. We enforce a *grammar-first fallback*: semantic masking is skipped and the grammar layer alone controls selection. Structural integrity is a hard constraint; type violations are soft errors detectable by downstream validation, with DR (ontologically invalid type pair) more severe than CF (individually valid triples with cross-triple type inconsistency). Extraction tasks are robust to format-level constraints (Tam et al., 2024), supporting this asymmetric treatment.

Constraint Tightness. Per-relation constraint tightness $\kappa(r) = 1 - |\sigma(r)|/|\mathcal{E}|^2$, with schema mean $\bar{\kappa}$. Across WikiEvents ($\bar{\kappa}$=0.029), SciERC ($\bar{\kappa}$=0.333), DocRED ($\bar{\kappa}$=0.842), DR reduction ranges from 94–99% to 42–90%; this correlation is descriptive (n=3, p=0.138); entity type confusability is a critical co-factor (Appendix N, M); high-$\bar{\kappa}$ relations account for only 3.8% of gold triples in DocRED (Appendix N).

Greedy limitation and fallback cascade. The semantic layer commits entity types greedily—an instance of Level 3 (non-local) constraints (§3)—so CF violations arise when the same entity receives different types across triples; beam search mitigates this (§5.4). Fallback triggers on 0.5–5.7% of type-bearing positions (zero-shot), virtually eliminated by fine-tuning ($\leq$0.11%); all residual DR occurs at fallback positions (Appendix C).

Non-rectangularity and independent enforcement. Type signatures are non-rectangular (0/7 SciERC, 23/96 DocRED rectangular); independent enforcement admits 33–37% spurious pairs, eliminated by fine-tuning (Table 2). Shuffling type order produces byte-identical outputs across all 551 SciERC documents, confirming order invariance (Appendix N).

	Violation Rates (%)						F1	
Mode	DR↓	ΔDR	RV↓	ET↓	CF	Tot.↓	Rel.↑	Ent.↑
Unconstrained	15.37	—	1.39	0.40	0.08	15.45	8.23	31.01
Grammar	14.84	↓3%	**0**	**0**	0.08	14.92	8.23	30.98
GCD-equiv.†	15.26	↓1%	**0**	**0**	0.04	15.30	8.75	30.91
Semantic	11.65	↓24%	1.51	0.40	2.71	14.36	8.15	31.01
S1	**0.93**	**↓94%**	**0**	**0**	9.90	**10.83**	**8.36**	30.98
S2	0.97	↓94%	**0**	**0**	8.03	**9.00**	8.03	**31.90**

Table 1: Five-way ablation on SciERC (551 docs, Qwen3-8B, zero-shot). ΔDR: relative change vs. unconstrained; Tot. = DR + CF; bold = best (CF excluded). F1 ×100. S1/S2 = single/two-pass (§3.4). Grammar = GCD ceiling (Geng et al., 2023). †vLLM `guided_json` (Kwon et al., 2023) (§5.4).

5 Experiments

We evaluate OntoCoDec in two deployment configurations: greedy S1 (DR $\leq$ 0.09%, +10% latency) and beam S1 (all violations $\rightarrow$ 0%, 1.54× latency); zero-shot evaluation (§5.1–§5.3) serves as a diagnostic probe for the fine-tuned setting (R-F1 41–47%).

Setup. We evaluate on SciERC (Luan et al., 2018) (551 test sentences, 7 relations, $\bar{\kappa}$=0.333) and DocRED (Yao et al., 2019) (998 dev documents,[1] 96 relations, $\bar{\kappa}$=0.842). RE type constraints are extracted from observed type pairs in training data (neither dataset provides a formal ontology); EE constraints on WikiEvents (§5.4) derive from the KAIROS ontology. All constraints are schema-level, independent of evaluation splits. Primary model: Qwen3-8B (Qwen Team, 2025), zero-shot greedy decoding; cross-model validation in §5.3, composability with fine-tuning in §5.1. LoRA hyperparameters were selected on SciERC development data and transferred to DocRED without further tuning; DocRED development documents were used for evaluation only. Four violation categories: **DR** (type pair $\notin \sigma(r)$), **RV** (relation $\notin \mathcal{R}$), **ET** (entity type $\notin \mathcal{E}$), **CF** (conflicting types across triples). Baseline DR: 15.37% (SciERC), 18.46% (DocRED); schema-informed prompting reduces DR by 62% but is per-relation unreliable (Appendix C).

5.1 Ablation and Composability with Fine-Tuning

Table 1 presents the five-way ablation on SciERC (Qwen3-8B, zero-shot, 551 documents); only the active constraint layers differ across rows.

[1] DocRED test annotations are not publicly available; we evaluate on the development set.

Zero-shot diagnostic. S1 reduces DR from 15.37% to 0.93% (↓94%) with unchanged R-F1 (p=0.58), confirming the *empirical enabling dependency*: grammar-only eliminates RV/ET but achieves only 3% DR reduction; semantic-only reaches 24%; their composition yields 94% (Table 1). Total type-violation burden decreases from 15.45% to 10.83%, reflecting *violation conversion* (CF 0.08% → 9.90%) that fine-tuning resolves (Table 2). S2 achieves comparable DR but lower R-F1; we adopt S1 (per-relation patterns in Appendix C). This DR-to-CF conversion represents a severity downgrade: DR violations introduce ontologically invalid type pairs (irrecoverable without re-extraction), while CF violations—individually valid triples with cross-triple inconsistency—are amenable to post-hoc resolution (TCR eliminates 18–92%; Appendix M).

Composability with fine-tuning. Fine-tuning alone reduces DR to 1–2% but leaves residual violations on tight-constraint relations (Table 2); adding S1 yields DR ≤ 0.09% with no significant ΔR-F1 after Bonferroni correction (min. p=0.014; 95% CIs ⊆ $[-0.76, +0.22]$ pp; removed triples are 97.6–100% false positives; ≤1.1% of true positives are affected across four fine-tuned configurations, with recall cost ≤0.51 pp; Appendix J.5). Under fine-tuning, grammar becomes a no-op (RV/ET already 0%; ΔDR<0.2 pp), making the semantic layer the sole DR defense (39–69% independent reduction; Appendix C). On SciERC, three relations with $\kappa \geq 0.36$ retain DR under fine-tuning alone; S1 eliminates all residual violations (Appendix C). CF drops from +9.82 pp (zero-shot) to +1.18 pp (fine-tuned), concentrated in GENERIC↔X ambiguity (63.3%; Appendix C); beam (k=4) eliminates all violation types (R-F1 43–47; §5.4).

Downstream impact. Among constraint-removed triples, only 0.49% match gold labels (Appendix C); error analysis of 514 false negatives confirms only one (0.2%) was blocked by constraints, with dominant sources (relation confusion, entity type polysemy) schema-valid and beyond constraint reach (Appendix J.5). Across fine-tuned settings, 36 residual DR violations decompose into entity type errors (39%), schema coverage gaps (33%), and relation/direction errors (28%); the irreducible floor is ∼3 violations (Appendix C).

Model	Dataset	Mode	DR↓	CF	R-F1↑	E-F1↑	ΔDR
Qwen3-8B	SciERC	Unconstr.	1.95	0	43.86	67.05	—
		S1	**0**	1.18	43.77	67.05	↓100%
	DocRED	Unconstr.	0.56	0.08	40.89	82.35	—
		S1	**0.00**	0.22	40.79	81.87	↓100%
Llama 3.1-8B	SciERC	Unconstr.	1.11	0	45.45	67.76	—
		S1	0.09	0.46	45.47	67.76	↓92%
	DocRED	Unconstr.	0.49	0.08	44.55	84.77	—
		S1	0.00	0.26	44.70	84.71	↓100%

Table 2: Fine-tuning composability: LoRA × unconstrained vs. S1 on SciERC (551 docs) and DocRED (998 dev docs). F1 ×100. No ΔR-F1 is significant (Bonferroni-corrected; details in text); all DR reductions are (p<0.001). E-F1 includes parse-failure documents (Qwen PF: 7–8%; Llama PF: 2–3%).

5.2 Cross-Dataset Validation on DocRED

DocRED (96 relations, 14× SciERC scale) validates the pattern under a larger, more restrictive ontology (Table 12). S1 reduces DR from 18.46% to 10.21% (↓45%), eliminates RV (29.66% → 0.03%) and ET, with R-F1 +0.50 pp ($p < 0.001$). The lower DR reduction (45% vs. 94% on SciERC) reflects DocRED's tighter constraints, which trigger more grammar-first fallback (§4). Grammar-only *increases* DR (18.46% → 26.17%) by exposing latent violations when OOV relations become evaluable—a distinct cross-dataset confirmation of the enabling dependency. Fine-tuning resolves residual DR entirely (Table 2; residual decomposition in Appendix C).

5.3 Cross-Model Validation

Table 3 replicates S1 with Llama-3.1-8B and Mistral-7B under identical prompts.

Results. RV and ET reach ≤0.03% for all models on both datasets. DR drops 76–99% on SciERC and 42–90% on DocRED; R-F1 improves in 5/6 settings. CF direction depends on calibration: S1 introduces CF for calibrated models (up to 12.02%; §7) but reduces it for Mistral (56.87% → 1.27%); fine-tuning resolves both (§5.1). Post-hoc filtering achieves DR=0% but at 15–74% rejection rates (Appendix C).

5.4 Extension to Event Extraction (Preliminary)

We apply OntoCoDec to event extraction (EE) on WikiEvents (Li et al., 2021) (56 event types, 80 roles, 27 entity types, 245 constraints; Appendix C) without modification. Violations extend to EV (event vocabulary), RoV (role vocabulary), and

Dataset	Model	Violation Rates (%) DR↓	RV↓	ET↓	CF	R-F1 ↑
SciERC	Qwen (U)	15.37	1.39	0.40	0.08	8.23
	Qwen (S1)	0.93	0	0	9.90	8.36
	Llama (U)	13.34	0.55	0	0.24	2.68
	Llama (S1)	0.08	0	0	6.56	2.54
	Mistral (U)	56.94	0	1.50	56.87†	3.66
	Mistral (S1)	13.40	0	0	1.27	3.67
DocRED	Qwen (U)	18.46	29.66	2.48	0.10	7.74
	Qwen (S1)	10.21	0.03	0	2.57	8.23
	Llama (U)	26.07	16.03	0.13	0.17	5.98
	Llama (S1)	2.55	0	0	12.02	6.27
	Mistral (U)	57.42	42.35	7.58	24.78	4.34
	Mistral (S1)	33.20	0	0	0.19	5.52

Table 3: Three-model zero-shot validation. U: unconstrained; S1: full two-layer. Parse failures: Llama DocRED 9.1%; others ≤1.1%. †Mistral: poor zero-shot calibration causes pervasive type inconsistencies. F1 ×100.

AC (argument constraint, analogous to DR).

Results. Unconstrained EE produces extreme vocabulary violations (EV+RoV >85%; Table 4), with entity F1 = 0 because the parser cannot process off-schema output. S1 eliminates EV and RoV across all models, which validates the tokenizer-agnostic design across byte-level BPE and SentencePiece. Grammar-only eliminates EV but leaves RoV at 64–77% and surfaces latent AC (8–14%), consistent with DocRED's pattern (§5.2); S1 drives RoV to 0%. Residual AC (1.3–3.8%) traces to semantically similar types (e.g., SID vs. SEN). CF rises to 15–21%, exceeding RE levels (10–12%) because EE's multi-argument structures increase entity reuse across slots. Post-hoc filtering is impractical (<0.5% arguments retained). Unlike RE, EE shows measurable F1 gains: entity F1 0% → 12–18%, trigger F1 +1.5–1.8 pp.

Composability with fine-tuning (preliminary, n=20). On the full official test split (20 documents; 194/20/20 (Li et al., 2021)), LoRA + S1 eliminates RoV (Qwen 38.31% → 0%, Llama 8.8% → 0%) and reduces AC to ≤0.8%, with argument F1 within ±1.1 pp; trigger F1 shifts (−5.4/−0.6 pp) have wide CIs consistent with n=20. CF remains low (0.77–1.18%); per-document variance is high (Arg-F1 std/mean >0.5), confirming sufficient power to detect violation elimination but not small F1 differences (power analysis in Appendix G).

Model	Mode	Violation Rates (%) EV↓	RoV↓	AC	CF	F1 Trig.↑	Ent.↑
Qwen3-8B	Unconstr.	24.0	74.9	0.3	0	11.8	0
	Grammar	**0**	76.1	8.2	—	—	—
	S1	**0**	**0**	2.1	21.1	13.4	14.1
Llama 3.1-8B	Unconstr.	10.5	88.8	0.1	0	5.7	0
	Grammar	**0**	77.0	8.8	—	—	—
	S1	**0**	**0**	3.8	17.2	7.5	17.6
Mistral 7B	Unconstr.	37.3	63.9	0	0	2.4	0
	Grammar	**0**	64.0	13.8	—	—	—
	S1	**0**	**0**	1.3	14.9	3.9	11.7

Table 4: WikiEvents EE results. Zero-shot rows (n=246): full WikiEvents corpus; fine-tuning (§5.4, n=20): standard 194/20/20 split (Li et al., 2021). Violation rates over parsed documents. — indicates not applicable (grammar-only has no semantic-layer metrics). F1 × 100. Fine-tuning results are preliminary given the small test set (n=20); bootstrap 95% CIs for argument F1 span approximately ±3–5 pp.

Runtime overhead. S1 overhead is +13.1% on SciERC, lower on DocRED due to amortization over longer sequences; under LoRA, +10.0% with 37% lower absolute latency. Constraint compilation is one-time <0.5 s; scaling benchmarks confirm constant overhead (2–3%) up to 200 entity types and 1,000 relations (Appendix K).

Beam search compatibility. With LoRA, beam search (k=4) (Anderson et al., 2017) eliminates all four violation types on both SciERC and DocRED at both 8B and 70B scales (R-F1 43–47; Appendix F), yielding two deployment tiers: greedy S1 (+10% latency, DR<0.1%, CF≤1.4%) and beam S1 (1.54× latency, all violations→0%; Table 29).

70B-class validation. On Llama-3.1-70B-Instruct (AWQ INT4), zero-shot S1 eliminates DR on both datasets: SciERC 6.41% → 0% (R-F1 11.24), DocRED 14.98% → 0% (R-F1 13.05); CF rises to 0.04–3.50%, comparable to 8B zero-shot CF. A four-way ablation on SciERC reveals scale-dependent layer interaction: semantic-only achieves 84% DR reduction (6.41% → 1.01%), far exceeding 8B (24%; Table 1); grammar remains necessary for RV elimination and CF control (semantic-only CF=76), suggesting grammar-constrained decoding is necessary but increasingly insufficient as models improve. LoRA confirms composability: DR → 0% on both datasets (R-F1 44.53, 43.95; $|\Delta| \leq 0.20$ pp; CF 0.18–1.33%). We note that 70B validation uses a single model family (Llama-3.1); cross-family generalization

at this scale remains to be verified. DocRED beam (k=4) S1 reduces parse failures from 12.5% to 0.5%, contributing to R-F1 +4.19 pp; the SciERC controlled comparison (0% parse failure in both settings) isolates pure beam quality gain at +0.28 pp (Appendix C).

Comparison with grammar-constrained approaches. Our grammar-only ablation (Table 1, row 2) implements the same constraints as GCD (Geng et al., 2023) and BoostCD (Sakota and West, 2025); vLLM `guided_json` (Kwon et al., 2023) confirms equivalence (DR=15.26%, R-F1=8.75; GCD-equiv. row, Appendix C). The ∼15% residual DR is the gap our semantic layer addresses (94% total reduction). Trie-only constraints without JSON enforcement yield 9–51% parse failure (Appendix C).

6 Related Work

We provide a detailed survey in Appendix B; here we position OntoCoDec along three axes.

Constrained decoding. XGrammar (Dong et al., 2025), Outlines (Willard and Louf, 2023), SynCode (Ugare et al., 2024), SGLang (Zheng et al., 2024), and related systems (Beurer-Kellner et al., 2023; Park et al., 2024; Geng et al., 2025) enforce format validity via CFG-based token masking. NeuroLogic Decoding (Lu et al., 2021) enforces predicate-logic constraints via beam-level reranking; OntoCoDec applies constraints per-token, avoiding beam-width dependence. In code generation, PICARD (Scholak et al., 2021) and Mündler et al. (2025) compile type systems into prefix automata; unlike PICARD's rejection-based pruning, OntoCoDec applies constructive forward masking via a dynamic entity registry (§4). DCCD (Reddy et al., 2026) and Nguyen et al. (2026) decouple reasoning from constrained generation via multi-pass architectures; OntoCoDec enforces schema semantics within a single forward pass.

Schema-aware IE. GoLLIE (Sainz et al., 2024), KnowCoder (Li et al., 2024), GUIDEX (Fuente et al., 2025), GenIE (Josifoski et al., 2022), TANL (Paolini et al., 2021), REBEL (Huguet Cabot and Navigli, 2021), InstructUIE (Wang et al., 2023), and others (Fei et al., 2022; Wang et al., 2022; Lu et al., 2022; Lou et al., 2023; Zhou et al., 2024) encode ontology structure or frame IE as structured generation, though schema compliance gaps persist (Wadhwa et al., 2023). GoLLIE and KnowCoder embed schema knowledge via code-style prompts (training-time internalization without runtime enforcement); GoLLIE-7B eliminates DR at severe coverage cost (R-F1=0.0, 46% empty output), while OntoCoDec achieves R-F1=45.47 at DR=0.09% (Li et al., 2024). ADELIE (Qi et al., 2024) aligns LLMs on IE through schema-aware instruction tuning but relies on model internalization rather than decoding-time enforcement, offering no structural guarantee against violations.

Constraint enforcement in IE. Roth and Yih (2004) and Riedel et al. (2010) formulated hard type constraints via ILP; post-hoc validation (Khorshidi et al., 2025; Caufield et al., 2024; Cotti et al., 2025) detects violations after generation but discards up to 48% of predictions. EnriCo (Zaratiana et al., 2024) uses ASP; IterGen (Ugare et al., 2025) backtracks on violations—OntoCoDec uses forward-only masking at $O(|V|)$ per step. Grammar-only enforcement (GCD (Geng et al., 2023), BoostCD (Sakota and West, 2025)) achieves only 3% DR reduction (§5.4); extraction tasks are robust to format constraints (Tam et al., 2024) despite degradation in other domains (Banerjee et al., 2025; Lee et al., 2026).

7 Discussion and Conclusion

OntoCoDec translates IE ontologies into composable token-level constraints, automatically compiled from ontologies and validated across 3 datasets, 3+ models, and 2 tasks. LoRA + greedy S1 near-eliminates DR ($<0.1\%$) at R-F1 41–47%; beam (k=4) eliminates all violations at 1.54× latency. The central insight is that IE type constraints exceed context-free expressiveness (Proposition 1): the semantic layer applies constructive forward masking via a dynamic entity registry, bridging the gap grammar-only enforcement cannot address (3% vs. 94% DR reduction). This enabling dependency is scale-sensitive (§5.4), suggesting grammar-constrained decoding is necessary but increasingly insufficient as models improve. Extending to global entity-level type consensus during decoding—eliminating CF without beam search overhead—is the primary direction for future work.

Limitations

Constraint scope. CF impact is model-dependent: for well-calibrated models (Qwen, Llama; DR $\leq 1\%$ under greedy constrained decoding), S1 introduces CF violations (up to 12.02%) as a consequence of per-relation independent enforcement; for poorly-calibrated models (Mistral), S1 substantially reduces CF ($56.87\% \rightarrow 1.27\%$) by constraining type assignments to schema-valid values. CF fallback ablation shows that eliminating CF re-routes violations to DR (+13.85 pp). CF↔DR thus represents a fundamental trade-off in composable constraint systems rather than an independently resolvable limitation (Appendix C). Global entity-level type consensus during decoding offers the most direct path to resolving CF violations. CF type inconsistencies may propagate to downstream tasks that consume extracted triples (e.g., knowledge graph construction, question answering); the impact of such propagation has not been evaluated. Entity F1 impact is model-dependent (+1.22 pp for Qwen vs. −0.64 pp for Llama on DocRED), partially confounded by increased parse failures under constraints (Llama: 9.1% vs. 6.8%). On 70B DocRED LoRA, parse failure rises modestly under S1 ($10.0\% \rightarrow 12.5\%$) due to generation timeouts on long documents rather than JSON format errors; SciERC (shorter documents) shows 0% parse failure in both settings. Residual DR under fine-tuning (36 violations across all fine-tuned settings, both models and both datasets) decomposes into entity type errors (39%), schema coverage gaps (33%), and model generation errors correctly caught by constraints (28%); Llama DocRED alone accounts for 33 of 36 residual violations (92%), reflecting weaker entity typing (Appendix C). Schema coverage gaps (one-third of residual violations) arise because RE constraints are extracted from training-data type pairs, which may not cover all valid combinations; larger training sets or manual constraint table completion can reduce this gap. Post-hoc type consensus re-voting (Appendix M) reduces CF violations by 18–92% depending on model and dataset, confirming that first-assignment ordering sensitivity is a primary CF source; however, integrating consensus into the decoding loop remains future work.

Evaluation coverage. Zero-shot results cover three 7–8B-class models, where relation F1 remains around 4–8% compared to ∼45% for fine-tuned systems. Zero-shot evaluation covers three 8B-class and one 70B-class model; fine-tuning composability is validated on 8B-class (two models, full datasets) and 70B-class models (one model; SciERC full, DocRED 200-document subset). Scaling to smaller models has not been verified; the observed relationship between schema restrictiveness and enforcement difficulty is based on only three schemas. Composability with fine-tuned models is validated on SciERC, DocRED, and WikiEvents with Qwen3-8B and Llama-3.1-8B (RE: DR $\leq 0.09\%$, R-F1 within ± 0.8 pp; EE: RoV→0%, AC≤0.8%), but evaluation is limited to one fine-tuning method (LoRA); residual CF (Qwen 1.18%, Llama 0.46% on SciERC) traces to schema-level type ambiguity (GENERIC↔X) rather than enforcement failure. WikiEvents fine-tuning composability is preliminary (20-document test set, the full official split). All main experiments use greedy decoding; our evaluation of beam search on the full SciERC (551 documents) and DocRED (998 documents) test sets across three models confirms compatibility (Appendices E, F), but comprehensive evaluation across all decoding configurations remains future work.

Engineering considerations. Scalability benchmarks up to 200 entity types and 1,000 relations show constant 2–3% inference overhead and sub-second compilation (<0.5 s; Appendix K); however, the first-token uniqueness requirement for continuation masking may become a tokenizer-dependent bottleneck beyond this scale. Our approach requires access to model logits for token-level masking, which excludes API-only models (e.g., GPT-4, Claude) where only output text is available; extending to API-only settings would require alternative enforcement mechanisms such as constrained sampling APIs or output reranking. Continuation masking assumes each entity type name maps to a distinguishable token prefix sequence. This assumption holds for our evaluation ontologies but may be challenged by ontologies with morphologically similar type names (e.g., `Protein` vs. `Protein_Complex` vs. `Protein_Domain`); when type names share a common prefix longer than one token, masking granularity depends on the tokenizer's subword segmentation. 70B experiments use AWQ INT4 quantization, which may alter logit distributions at mask boundaries; the impact on constraint enforcement precision has not been isolated. Our

evaluation is limited to English-language datasets. The grammar layer (CFG compilation via XGrammar) is tokenizer-agnostic, applying unchanged to any BPE or SentencePiece tokenizer including those trained on CJK-heavy corpora. The semantic layer's continuation masking is likewise language-independent in mechanism: prefix matching over type name tokens operates identically regardless of script, though CJK entity names produce more fragmented tokenizations under byte-level BPE (2–3$\times$ more tokens per entity than English equivalents), increasing the number of masking steps per type field without affecting correctness. Validation on Chinese IE benchmarks (e.g., DuIE, CLUENER) remains future work. Fine-tuning composability was validated exclusively with LoRA; full fine-tuning with an untuned learning rate underperforms LoRA (R-F1 37.89 vs. 43.77 on SciERC) but confirms that constraints remain effective (DR 2.74% $\rightarrow$ 0%; Appendix L); other fine-tuning methods may exhibit different constraint interaction patterns and warrant further investigation. The two-pass strategy (S2) exhibits scale-dependent degradation: at 70B, S2 reduces R-F1 by 8.45 pp while 8B models show comparable S1/S2 performance, suggesting optimal constraint granularity varies with model capacity (Appendix C).

Ethics Statement

This work uses publicly available benchmark datasets (SciERC, DocRED, and WikiEvents) with no personally identifiable information. No human subjects were involved in the experiments. The constrained decoding method operates on model outputs and does not introduce new privacy or safety risks beyond those inherent in the base language models. Code and data will be released upon acceptance.

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A Prompt Templates

All experiments use the same prompts across constrained and unconstrained decoding; constraints are applied at the logit level, not through prompt modification. Table 5 shows the system and user prompts for each dataset.

B Extended Related Work

Grammar-Constrained Decoding. Recent work has established efficient algorithms for restricting LLM outputs to conform to context-free grammars, making grammar-constrained decoding the de facto standard for obtaining structured output from language models. XGrammar (Dong et al., 2025) partitions the vocabulary into context-independent and context-dependent tokens and maintains persistent pushdown automata across decoding steps, achieving up to 100× speedup over prior CFG-constrained decoding methods. Outlines (Willard and Louf, 2023) reformulates constrained generation as finite-state machine transitions with offline vocabulary indexing, which enables efficient enforcement of regular-expression and JSON-schema constraints without runtime overhead per token. LMQL (Beurer-Kellner et al., 2023) introduces a declarative query language that compiles type, length, and regex constraints into token-level masks, providing a programmable interface for specifying multi-variable output structures. These systems ensure that every generated token belongs to a grammar-valid continuation, but they do not address the *distribution distortion* introduced by naive token masking. A parallel line of work tackles this problem directly: DOMINO (Beurer-Kellner et al., 2024) minimizes distributional divergence during CFG enforcement by formulating mask selection as an optimization problem, and ASAp (Park et al., 2024) provides a provably unbiased sampling algorithm that preserves the original LM distribution under grammar constraints. Together, these advances have made grammar-constrained decoding both fast and statistically principled, and provide a robust foundation for structured generation. However, all of these methods operate exclusively at the syntactic level: they ensure format validity (e.g., well-formed JSON, valid enum values) but cannot enforce cross-field semantic constraints such as ontology domain/range rules that depend on previously generated content. This syntactic-only limitation motivates OntoCoDec's semantic extension: our grammar layer builds on XGrammar for structural (JSON schema and enum) constraints, while a second semantic layer enforces relational type constraints that are inherently beyond the expressive

SciERC — System Prompt (Unconstrained & Constrained)

```
You are an expert information extraction system for scientific texts. Extract
entities and relations from the given text.
Entity types: Task, Method, Material, Metric, OtherScientificTerm, Generic
Relation types: Used-for, Feature-of, Hyponym-of, Part-of, Compare,
Conjunction, Evaluate-for
Output a JSON object with:
- "entities": list of {"text": "...", "type": "..."}
- "relations": list of {"head": "...", "tail": "...", "type": "...",
"head_type": "...", "tail_type": "..."}
Output ONLY valid JSON, no explanation.
```

SciERC — User Prompt

```
Extract all entities and relations from this scientific text:
"{text}"
Output JSON:
```

DocRED — System Prompt (Unconstrained & Constrained)

```
You are an information extraction system. Extract all entities and relations
from the following document.
Entity types: PER, ORG, LOC, TIME, NUM, MISC
Relation types: country, instance of, performer, ... (96 Wikidata relation
types; full list omitted for space)
Output a JSON object with exactly this structure:
{"entities": [{"text": "entity name", "type": "ENTITY_TYPE"}],
 "relations": [{"head": "head entity", "tail": "tail entity", "type":
"relation name",
  "head_type": "ENTITY_TYPE", "tail_type": "ENTITY_TYPE"}]}
Output ONLY the JSON, no other text.
```

DocRED — User Prompt

```
(Document text provided inline in the user message.)
```

Table 5: Extraction prompts used in all experiments. The same prompts are used for both unconstrained and constrained decoding; OntoCoDec's constraints operate at the logit level during generation. Schema-informed prompts (§5) additionally include valid (head type, tail type) pairs per relation.

power of context-free grammars.

Semantic Constrained Decoding for Code. In programming-language generation, constrained decoding has been extended beyond syntax to enforce semantic properties such as type correctness; this work demonstrates that richer-than-CFG constraints can be compiled into token-level filtering. In the text-to-SQL domain, PICARD (Scholak et al., 2021) maintains an incremental SQL parsing state that rejects tokens violating syntax or referencing non-existent database columns at each decoding step, achieving state-of-the-art results on Spider and CoSQL without modifying model weights. While mechanistically similar to OntoCoDec's token-level filtering, PICARD operates over a *fixed, fully enumerated* schema (database table and column names known before decoding begins). In contrast, OntoCoDec's entity registry is populated *during* generation as entity spans are extracted from free text; constraint sets evolve with the decoding prefix, a fundamental distinction between querying a known schema and extracting from unstructured text. Synchromesh (Poesia et al., 2022) pioneers Constrained Semantic Decoding (CSD), which enforces syntax, type, and scope constraints simultaneously during code generation from pre-trained LMs without fine-tuning; this establishes that semantic constraints can be expressed as dynamic token masks conditioned on the generation prefix. MGD (Agrawal et al., 2023) uses static-analysis monitors to perform incremental type checking at decode time; SantaCoder-1.1B with MGD outperforms text-davinci-003 on repository-level code completion by ensuring that every generated identifier is type-correct with respect to the surrounding code context. Most recently, Mündler et al. (2025) compile full type systems, including generic types and subtyping, into prefix automata for token-level logits masking and reduce compilation errors by over 50%. This last system is the closest mechanistic analog to OntoCoDec, applying the same "type system → automaton → logits mask" pipeline,

though in a fundamentally different domain. The key technical difference lies in the constraint structure: programming-language type systems feature deep subtyping hierarchies and parametric polymorphism, whereas IE ontology axioms specify flat domain/range sets; a relation R accepts head entities from set $\mathcal{D}(R)$ and tail entities from set $\mathcal{R}(R)$, with no subtype chains. Conversely, IE introduces a challenge absent in code generation: the constraint target (entity type) must be inferred from free-text spans rather than read from declared type annotations, so a stateful entity registry must track extracted types during autoregressive generation. These methods collectively demonstrate that semantic (type) constraints at decode time are both feasible and highly effective, but exclusively in the code domain where type systems are formally specified. OntoCoDec is the first to *automatically compile* IE ontology axioms, including domain/range constraints, into token-level decoding masks for information extraction, migrating the type-constrained decoding paradigm from code generation to IE.

Vocabulary and Type-Constrained Generation. The idea of enforcing entity-type compatibility constraints for information extraction dates back to pre-neural methods. Roth and Yih (2004) formulate joint entity and relation identification as an integer linear program (ILP) that enforces type constraints (e.g., EMP-ORG only between PER and ORG entities); this establishes the foundational principle that relational type compatibility should be a hard constraint, not a soft preference learned from data. In the autoregressive era, GENRE (De Cao et al., 2021) pioneers trie-based constrained beam search for autoregressive entity retrieval, restricting each decoding step to tokens consistent with valid entity names from a predefined catalog. GenIE (Josifoski et al., 2022) extends this approach to end-to-end information extraction with state-based constraints that enforce valid triple structures, achieving strong performance on closed-world knowledge base population, though it requires task-specific fine-tuning on the target schema. AdaKGC (Ye et al., 2023) introduces schema-constrained dynamic decoding with trie-based prefix trees that adapt to the target schema, enforcing that generated entity and relation names belong to a predefined inventory, a vocabulary-level constraint that ensures name validity but does not verify type compatibility between a relation and its arguments. GCD (Geng et al., 2023) provides a unified grammar-constrained decoding framework for structured NLP tasks including IE, introducing input-dependent grammars that adapt constraints based on the source text. GCD is the closest precursor to OntoCoDec in applying logits masking to IE, though limited to syntactic constraints. NeuroLogic Decoding (Lu et al., 2021) expresses lexical inclusion and exclusion constraints as predicate logic formulas enforced during beam search, offering a general mechanism for token-level control that can encode vocabulary membership but not relational type rules. These approaches constrain *which tokens* can appear (vocabulary membership) but not *which type combinations* are valid for a given relation. OntoCoDec's semantic layer enforces relational constraints (e.g., that the head entity of relation R must belong to type set $\mathcal{D}(R)$) which cannot be expressed as static vocabulary restrictions because the valid set depends on the relation type generated earlier in the same sequence. While Roth & Yih's ILP formulation shares the same type-compatibility motivation, it operates as a post-hoc reranker over discrete candidates; OntoCoDec integrates enforcement directly into autoregressive decoding, preventing violations rather than filtering them.

Structural Constraints via Training and Prompting. An orthogonal approach encodes IE output structure through model training or prompt design rather than decoding-time enforcement, relying on learned representations to produce schema-compliant outputs. UIE (Lu et al., 2022) defines a Structured Extraction Language and trains a unified text-to-structure model across 13 IE datasets, relying on large-scale pre-training to internalize output format constraints. GoLLIE (Sainz et al., 2024) expresses annotation guidelines as Python dataclass definitions and fine-tunes Code-LLaMA to generate typed code, achieving strong zero-shot transfer to unseen IE schemas through the implicit type structure of programming languages. KnowCoder (Li et al., 2024) takes this further by encoding 29K entity types from Wikidata as a Python class hierarchy, training a Code-LLM to generate structured extraction code that implicitly respects type inheritance, the most comprehensive schema-as-code approach to date. CodeIE (Li et al., 2023) similarly reformulates IE as code generation, using the structural regularity of programming languages to reduce output format errors. ADELIE (Qi et al., 2024) uses

SFT and DPO to align LLMs on IE tasks, achieving state-of-the-art results among open-source models while preserving general capabilities, though compliance with ontological constraints remains a learned soft preference rather than a hard constraint. In the supervised setting, joint entity-relation architectures such as SpERT (Eberts and Ulges, 2020) and PFN (Yan et al., 2021) enforce type consistency implicitly through shared representations: SpERT classifies entity spans and relation pairs in a single BERT pass with span-level negative sampling, while PFN decomposes feature encoding into entity-specific, relation-specific, and shared partitions via gating mechanisms. These models achieve strong performance but are limited to fixed, predefined schemas and cannot generalize to unseen ontologies without retraining, precisely the ontology-portable scenario requiring no constraint-specific training that OntoCoDec targets. EnriCo (Zaratiana et al., 2024) applies Answer Set Programming (ASP) constraints during decoding of joint entity-relation extraction models, enforcing structural type compatibility at generation time. This is a decoding-time approach, though with model-specific, manually authored constraints rather than ontology-driven compilation. Training-based methods provide *soft* structural guidance: the model learns to prefer valid outputs but cannot enforce compliance at inference time, as demonstrated by non-trivial violation rates even in fine-tuned systems. Importantly, training-based and decoding-time approaches are complementary rather than competing: a system like GoLLIE could be combined with OntoCoDec's constraint layers, using fine-tuning to maximize the probability of valid outputs while decoding-time enforcement eliminates residual violations. OntoCoDec provides *hard* decoding-time enforcement that requires no constraint-specific training, is automatically compiled from IE ontologies, and composable with any of these training-based approaches.

Post-hoc Ontology Validation. Rather than constraining generation, several methods validate LLM outputs against ontological rules after generation, applying a detect-then-fix paradigm. ODKE+ (Khorshidi et al., 2025) dynamically generates ontology snippets containing domain/range constraints as prompt context, achieving 98.8% precision across 195 predicates at web scale, but constraint adherence relies on prompt-level guidance, not enforcement, leaving residual violations unaddressed. SPIRES/OntoGPT (Caufield et al., 2024) provides an ontology-based extraction framework that grounds LLM outputs against OBO ontologies post-generation: it validates extracted entities and relations through ontology lookup and corrects mismatches in a second pass, the prompt-then-validate paradigm in the biomedical domain. OntoLogX (Cotti et al., 2025) applies SHACL validation to LLM-generated knowledge graphs, iteratively re-prompting the LLM to correct constraint violations and reducing them by approximately 10× across multiple correction rounds. This demonstrates the detect-then-fix paradigm's effectiveness but also its inherent multi-pass cost. Upadhyay et al. (2023) inject entity-type knowledge as constraints into Open IE via both constrained inference and constraint-aware training, improving F1 by 29% on CaRB, though their constrained inference operates over a discrete candidate set rather than during autoregressive generation. Regino and dos Reis (2025) propose a multi-stage LLM pipeline for validating RDF triple insertions against RDFS domain/range constraints, where each validation step requires a separate LLM call, effective but computationally expensive at scale. RELATE (Olasunkanmi et al., 2025) addresses the alignment gap between LLM-extracted biomedical relations and standardized ontology predicates through a three-stage pipeline: ontology preprocessing with predicate embeddings, similarity-based retrieval via SapBERT, and LLM-based reranking with negation handling, achieving 94% accuracy@10 on ChemProt. While RELATE demonstrates effective ontology grounding for relation extraction, its constraints are applied *post-extraction* through retrieval and reranking rather than during generation, and it requires domain-specific embedding models trained on biomedical corpora. OntoCoDec enforces ontology constraints *during* autoregressive decoding and requires no additional trained components beyond the base LLM. Concurrently, Zhou et al. (2026) compile OWL ontologies into executable tool interfaces within The World Avatar framework; LLM agents must invoke ontology-aware tools that enforce semantic constraints during knowledge graph construction from scientific text. This "ontology-to-tools" approach shares OntoCoDec's philosophy of compiling formal ontological knowledge into runtime enforcement mechanisms, but operates at the tool-call level (constraining which API functions the

agent invokes) rather than at the token level (constraining which tokens the decoder produces), and requires an agentic multi-step workflow rather than single-pass autoregressive generation. Mehenni and Zouaq (2024) introduce ontology-constrained beam scoring for clinical summarization, applying soft biases (not hard masks) during beam search to steer outputs toward ontology-compliant terms, a middle ground between unconstrained generation and hard enforcement that reduces but cannot eliminate violations. Post-hoc validation detects violations after generation and requires rejection or correction cycles that waste compute. In our setting, 15.37% of generated triples contain domain/range violations under unconstrained decoding, a substantial waste of generation effort. OntoCoDec *prevents* violations during decoding, redirecting probability mass to valid alternatives rather than discarding invalid outputs, with only marginal overhead from logits masking compared to the cost of regenerating or correcting violated triples. Soft-constraint approaches such as ontology-weighted beam scoring reduce violation frequency but cannot eliminate it, whereas OntoCoDec substantially reduces violations (DR↓94% on SciERC, ↓45–90% on DocRED depending on model).

Quality Impact of Constrained Decoding. A growing body of work raises concerns that constrained decoding degrades LLM generation quality, particularly for reasoning-intensive tasks where format enforcement interferes with chain-of-thought processes. Tam et al. (2024) conduct a comprehensive evaluation showing that rigid format constraints (JSON, XML) harm performance on math and multi-hop QA benchmarks but *benefit* classification-style tasks such as slot filling, a distinction directly relevant to IE, which is fundamentally an extraction (classification-like) task rather than a reasoning task. CRANE (Banerjee et al., 2025) provides theoretical analysis of why restricting outputs to narrow grammars suppresses chain-of-thought reasoning by eliminating intermediate reasoning tokens, and proposes an alternating strategy between unconstrained reasoning phases and constrained output phases to mitigate this effect. Schall and de Melo (2025) conduct a systematic study across 11 models and multiple benchmarks, finding 10–30% semantic correctness degradation on reasoning tasks under structural constraints, with larger models showing greater absolute degradation. Zhou (2026) identify "structure snowballing," a failure mode where format compliance cognitive load traps models in repetitive structural patterns rather than improving reasoning during self-correction iterations. These findings establish a legitimate concern for constrained decoding in reasoning-heavy applications. However, information extraction occupies a fundamentally different regime: the model extracts entities and relations from input text rather than generating novel reasoning chains, and Tam et al. (2024)'s own results show that extraction-style tasks are robust to, or even benefit from, format constraints. OntoCoDec addresses this concern empirically: domain/range violations decrease by 94% on SciERC (15.37%→0.93%) and 45–90% on DocRED across two models, with F1-neutral impact on SciERC (+0.13pp, p=0.58) and significant R-F1 gains on DocRED (Qwen $p<0.001$, Llama p=0.013). Critically, OntoCoDec's constraints target only type tokens (entity types, relation labels), leaving free-text spans (entity surface forms and their boundaries) entirely unconstrained, which minimizes the distribution distortion that causes quality degradation in reasoning tasks. A two-pass variant (S2), explored in ablation (Section 5), separates extraction and constraint enforcement into distinct passes, analogous to CRANE's alternating strategy; however, our primary single-pass approach already avoids reasoning interference by constraining only type tokens while leaving free-text spans unconstrained.

Our Approach. OntoCoDec occupies a unique position in this landscape: it is the first system to *automatically compile* IE ontologies, including domain/range axioms, into token-level semantic masks, extending grammar-constrained decoding from syntactic to semantic enforcement without manually authored rules. Unlike vocabulary-constrained methods that restrict individual tokens to a fixed set, OntoCoDec enforces *cross-field relational constraints* through a stateful entity registry that tracks extracted types during autoregressive generation; constraint evaluation is dynamic and depends on previously decoded content. This approach is orthogonal to supervised IE methods and applicable to autoregressive LMs with logits access without model modification or additional training (evaluated on 7B–70B models across three tokenizer families). Unlike training-based IE systems, OntoCoDec requires no constraint-specific training and substantially reduces type violations at inference time (DR↓94% on SciERC, ↓45–90%

on DocRED); unlike post-hoc validation, it prevents violations rather than detecting them, redirecting probability mass to valid alternatives with significant R-F1 gains on DocRED ($p \leq 0.013$) and F1-neutral impact on SciERC. The two constraint layers, grammar (structural) and semantic (relational), compose via bitwise AND with an empirical enabling dependency: the grammar layer ensures every generated type token belongs to the valid vocabulary, which enables the semantic layer to look up and enforce domain/range rules for that type.

C Extended Discussion

CF Dual Mechanism. CF violations arise from two distinct mechanisms: (1) *Registry override*: the semantic constraint forces a domain-compatible type that conflicts with the entity's previously registered type. The entity appears in multiple relations with different type requirements, and the per-relation mask selects the locally valid type without consulting prior assignments. (2) *Grammar-first fallback* (§3.3): when the semantic mask conflicts with all grammar-valid tokens, the semantic layer is dropped entirely, allowing the grammar layer to generate any structurally valid type, which may differ from the entity's registered type.

These mechanisms have sharply different model profiles. Qwen3-8B triggers zero grammar-first fallbacks across all experiments, attributing 100% of its CF violations to registry override. Llama-3.1-8B, by contrast, triggers 5,882 fallbacks on SciERC (10.7 per document) and 57,825 on DocRED (63.8 per document), making grammar-first fallback the dominant CF source. This asymmetry is consistent with the cross-model CF gap: Qwen achieves CF = 2.57% on DocRED vs. Llama's 12.02%, despite both using identical constraint configurations. The difference traces to Llama's less calibrated type predictions, which more frequently conflict with the semantic mask's valid set, triggering fallback.

Fallback Behavior and Effective Coverage. The two-layer architecture produces two distinct fallback mechanisms with different semantic consequences. *Grammar-first fallback* occurs when $S(t) \cap G(t) = \emptyset$: the semantic layer deactivates as a safety valve, preserving structural validity at the cost of potential type violations. The token-level fallback rate (36.6% on SciERC Llama) is inflated by continuation tokens of multi-token type names (e.g., OTHERSCIENTIFICTERM, averaging 2.87 BPE tokens); at semantically meaningful first-token positions, the rate is 0.04–5.7% (zero-shot) and drops to $\leq 0.11\%$ under fine-tuning. These events account for the residual DR that persists after constraint enforcement. *Registry-consistent fallback* (4.5%) occurs when the entity's registered type conflicts with domain/range constraints; the system defaults to the registered type for cross-triple consistency. Together, these mechanisms produce the characteristic profile of composable constraint systems: residual DR traces to grammar-first fallback events, while CF violations arise from registry override events where per-relation enforcement assigns conflicting types.

We define *effective semantic constraint coverage* as $1 - f_{\text{gf}}$, where f_{gf} is the grammar-first fallback rate. On SciERC, Llama-3.1-8B shows $f_{\text{gf}} = 36.6\%$ (6,237 of 17,024 mask calls), yielding 63.4% effective coverage. On DocRED, by contrast, Qwen3-8B shows $f_{\text{gf}} = 0.08\%$ (46 of 54,859), yielding 99.9% coverage. The large discrepancy is primarily a tokenization artifact: SciERC entity types are multi-token (e.g., OTHERSCIENTIFICTERM, averaging 2.87 mask calls per type field), and continuation tokens necessarily conflict with the semantic mask, inflating the per-mask-call fallback rate. DocRED types (PER, ORG, LOC, TIME, NUM, MISC) are all single-token, eliminating this inflation. This metric directly quantifies the empirical enabling dependency (§3.3): it measures the fraction of decoding steps where both constraint layers are jointly active, providing a fine-grained view beyond aggregate violation rates. Qwen3-8B, with zero grammar-first fallbacks, achieves 100% effective coverage on SciERC, consistent with its lower residual DR (0.93% vs. Llama's 0.08%).

Fallback Causal Tracing. To establish that grammar-first fallback is the *sole* mechanism producing residual DR violations, we instrument the constraint processor to log, for each violation, whether the semantic mask was active ($S(t) \cap G(t) \neq \emptyset$) or bypassed via fallback ($S(t) \cap G(t) = \emptyset$) at the type-bearing position. Table 6 reports the results across four SciERC configurations.

When the semantic mask is active, $\text{DR}_{\text{independent}} = 0$: the type constraint processor structurally prevents domain/range violations. All 18 residual DR violations across four configurations occur exclusively at fallback positions, confirming that grammar-first fallback is

Model	Setting	Fallback Rate	DR	From FB
Qwen3-8B	FT S1	0/2256 (0%)	0	N/A
Qwen3-8B	ZS S1	16/3458 (0.46%)	16	100%
Llama-3.1-8B	FT S1	1/2162 (0.05%)	1	100%
Llama-3.1-8B	ZS S1	1/2630 (0.04%)	1	100%

Table 6: Fallback causal tracing on SciERC (551 documents). Fallback Rate: type-bearing positions where $S(t) \cap G(t) = \emptyset$ triggers grammar-first fallback. DR: residual domain/range violations. From FB: fraction of DR violations occurring at fallback positions. All violations occur exclusively at tail_type positions (range constraints are tighter than domain constraints).

Configuration	DR (%)	CF (%)	DR+CF	f_{gf} (%)
S1 (baseline)	0.93	9.90	10.83	—
CF fallback	14.78	0.04	14.82	36.64
Δ	+13.85	−9.86	+3.99	—

Table 7: CF-DR trade-off pilot on SciERC (Qwen3-8B, 551 documents; `w1_cf_fallback_scierc_pilot`). CF fallback eliminates CF but re-routes violations to DR, increasing total violations (DR+CF) by 3.99 pp. f_{gf}: token-level grammar-first fallback rate at type-bearing positions (includes continuation tokens of multi-token type names; first-token position fallback rate is 0.46% for SciERC Qwen; §4).

a necessary condition for residual DR. Fine-tuning reduces fallback to near-zero ($\leq$0.05%), effectively eliminating the sole source of residual violations. All fallback events occur at tail_type positions, where $|\text{range}(r, \tau^h)|$ is smaller than $|\text{dom}(r)|$, making the intersection $S(t) \cap G(t)$ more likely to be empty.

Enabling Dependency: Confound Analysis. The empirical enabling dependency (§3) shows that the semantic layer achieves only 24% DR reduction without the grammar layer (Table 1). A potential confound is that the semantic layer depends not only on grammar-enforced vocabulary membership but also on structural enforcement for positional awareness: the semantic layer identifies type-bearing positions by pattern-matching the decoded prefix against the JSON schema, which is reliable only when the grammar layer guarantees valid JSON structure. Without grammar enforcement, malformed structure can cause the semantic layer to misidentify type-bearing positions, reducing effective coverage. We quantify this via the semantic-only ablation (Table 1, row 3): CF rises from 0.08% to 2.71%, indicating that 2.63 pp of additional violations trace to structural ambiguity. However, the dominant enabling mechanism remains vocabulary membership: the semantic layer's domain/range lookup requires $r_j \in \mathcal{R}$, which only the grammar layer ensures; without it, OOV relation names produce null constraint lookups. The confound reinforces rather than undermines the enabling dependency: both vocabulary and structural contributions of the grammar layer are prerequisites for effective semantic enforcement.

CF-DR Trade-off Analysis. CF violations are the most prominent residual limitation of OntoCoDec under zero-shot decoding (9.90% on SciERC, up to 12.02% on Llama DocRED). We treat this honestly: per-relation independent enforcement cannot eliminate CF without trading it against DR. Table 7 reports a pilot ablation in which we replace the grammar-first fallback with a CF-priority fallback that forces the entity's registered type when it conflicts with $\sigma(r)$.

The CF fallback variant nearly eliminates CF (9.90% $\rightarrow$ 0.04%, −9.86 pp) but at the cost of a 15.9$\times$ increase in DR (0.93% $\rightarrow$ 14.78%, +13.85 pp). This is not a net improvement: total violations (DR+CF) rise from 10.83% to 14.82% (+3.99 pp), so the change *re-routes* violations rather than reducing them. The mechanism is mechanical: each token position with conflicting masks must commit to one violation type; when the CF fallback forces the system to emit the entity's registered type even though it lies outside $\sigma(r)$, the resulting triple satisfies cross-entity consistency but violates domain/range. The token-level grammar-first fallback is triggered at 36.64% of type-bearing positions (0.46% at first-token positions; §4); this conflict is structural, not a tunable hyperparameter.

We interpret this as a fundamental tension in composable constraint systems with *per-relation independent* enforcement: enforcement priority can be allocated between DR and CF but cannot independently minimize both. Within this design, the current S1 configuration sits near the Pareto frontier of the trade-off, achieving 0.93% DR at the cost of 9.90% CF. The direction of the trade-off is consistent beyond SciERC/Qwen3-8B: across all six zero-shot configurations (3 models $\times$ 2 datasets), DR and CF are descriptively inversely associated, with no setting showing both high DR and high CF simultaneously (Table 3); however, n=6 is insufficient for statistical inference on this correla-

tion. The pattern is a structural consequence of per-relation enforcement with grammar-first fallback, not a model- or dataset-specific artifact; we select SciERC/Qwen3-8B for detailed ablation because it exhibits the highest CF (9.90%), making the trade-off most visible. While the inverse association holds across all configurations, the absolute trade-off magnitude may differ in settings with higher baseline CF, such as Llama DocRED (CF = 12.02%), where the larger CF headroom could yield proportionally larger shifts under alternative fallback priorities. The CF residual is concentrated in schema ambiguity rather than enforcement failure: 63.3% of CF violations on SciERC involve the catch-all GENERIC type (Table 9), which competes with every content type depending on anaphoric vs. referential use, a distinction the per-slot decoder cannot resolve without document-level coreference.

Two complementary paths can lower CF without re-routing it to DR. (i) *Reduce the baseline conflict rate.* Composing OntoCoDec with LoRA fine-tuning on SciERC reduces CF by 88% relative to the zero-shot setting (9.90% → 1.18%; Table 2), because the fine-tuned model's per-slot type predictions agree more often with the entity registry, so the per-relation mask and the registry intersect more frequently. The remaining 1.18% still traces to GENERIC↔X schema ambiguity, so this path is empirically validated on one dataset and one fine-tuning method but does not generalize without further evidence. (ii) *Replace per-relation enforcement with global entity-level type consensus during decoding.* This is outside the scope of OntoCoDec's forward-only autoregressive design and is left to future work.

Dual-Factor Quantitative Analysis. Figure 2 plots residual DR against the product $\bar{\kappa} \times \text{RV}_{\text{baseline}}$ for model×dataset configurations. For *calibrated* models (Qwen3-8B, Llama-3.1-8B; unconstrained ET<3% on both datasets), a linear fit yields $y = 0.366x - 0.138$ with $R^2 = 0.886$ (n=4, $F(1, 2) = 15.54$, $p \approx 0.058$); with only four data points (df = 2) and $p > 0.05$, this constitutes a descriptive characterization rather than a confirmatory statistical test. The strong fit supports the dual-factor pattern from §4: residual DR is governed by the product of schema tightness and baseline violation rate. Mistral-7B (unconstrained ET=7.58% on DocRED) deviates from this pattern: its residual DR on DocRED (33.21%) substantially exceeds

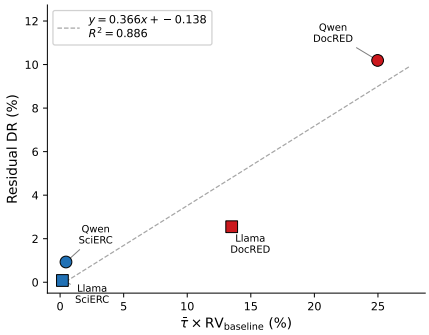


Figure 2: Residual DR vs. Schema Restrictiveness × Baseline Violation Rate ($\bar{\kappa} \times \text{RV}_{\text{baseline}}$) across four model×dataset configurations (n=3 schemas; descriptive only). Linear fit (calibrated models): $y = 0.366x - 0.138$, $R^2 = 0.886$, p=0.058 (n=4, df=2; not significant); Mistral deviates due to entity misclassification.

the linear prediction. When model calibration is insufficient (ET>5% on either dataset), residual DR is dominated by entity misclassification rather than constraint-search failure. This motivates a principled exclusion criterion: models with unconstrained ET>5% on either dataset fall outside the regime where the tightness×RV product is the primary DR driver. The two DocRED points (high $\bar{\kappa}$=0.842) separate cleanly from the two SciERC points (low $\bar{\kappa}$=0.333), and within each dataset the model with higher baseline RV shows higher residual DR.

Dataset-Dependent F1 Impact. The divergence in F1 significance between DocRED (+0.50 pp, $p < 0.001$) and SciERC (+0.13 pp, $p = 0.58$) traces to a single mechanism: grammar-layer elimination of out-of-vocabulary relation names. DocRED's baseline RV of 29.66% produces 5,189 OOV predictions counted as false positives; eliminating them yields a significant precision gain. SciERC's baseline RV is only 1.39% (24 OOV predictions), too few to produce a detectable F1 effect. Llama-3.1-8B's lower baseline RV (16.03%) yields a smaller but directionally consistent improvement (+0.29 pp vs. +0.50 pp); F1 gain magnitude is proportional to baseline OOV rate.

Bootstrap Confidence Intervals and Effect Sizes. Table 8 reports per-configuration bootstrap CIs (10,000 iterations, document-level resampling) for the fine-tuning composability results (Table 2).

Practical Significance of Zero-Shot F1. Our reported relation F1 of approximately 8% on both SciERC and DocRED reflects the inherent dif-

Model	Dataset	ΔR-F1 (pp)	95% CI	p
Qwen3-8B	SciERC	−0.10	[−0.30, 0.00]	0.726
Qwen3-8B	DocRED	−0.31	[−0.69, +0.01]	0.056
Llama-3.1-8B	SciERC	+0.02	[0.00, +0.07]	0.736
Llama-3.1-8B	DocRED	+0.06	[−0.11, +0.25]	0.535
DR% reduction (all $p < 0.001$):				
Qwen3-8B	SciERC	−1.95	[−2.87, −1.11]	<0.001
Qwen3-8B	DocRED	−0.75	[−1.07, −0.48]	<0.001
Llama-3.1-8B	SciERC	−1.02	[−1.98, −0.29]	<0.001
Llama-3.1-8B	DocRED	−0.29	[−0.53, −0.10]	<0.001

Table 8: Bootstrap CIs for Table 2 (10,000 iterations, document-level resampling, seed=42). Top: R-F1 difference (S1 − unconstrained); no comparison reaches significance at α=0.05 after Bonferroni correction (k=4, threshold p=0.0125). Bottom: DR% reduction; all significant. Effect sizes for R-F1 are negligible ($|\Delta| \leq$ 0.31 pp, all CIs spanning or adjacent to zero).

Conflicting Type Pair	Count	%
Generic ↔ OtherScientificTerm	85	34.7
Generic ↔ Material	35	14.3
Generic ↔ Metric	30	12.2
Method ↔ Task	20	8.2
Material ↔ Method	17	6.9
OtherScientificTerm ↔ Task	12	4.9
10 further pairs (each ≤3.3%)	46	18.8
Total	**245**	100.0

Table 9: CF violation distribution on SciERC S1 (Qwen3-8B, 551 documents). All 245 violations are pairwise conflicts. Conflicts involving GENERIC account for 63.3%.

ficulty of zero-shot closed IE, where the model receives no task-specific training examples. For reference, supervised state-of-the-art systems on SciERC achieve approximately 49–53% relation F1 with full training data (Zhong and Chen, 2021; Ye et al., 2022). Constraint enforcement operates largely independently of the extraction model's intrinsic capability: it restricts the output space to ontologically valid triples regardless of the underlying model's accuracy. Even after LoRA fine-tuning, residual DR of 1–2% persists on tight-constraint relations (Table 2); constraints remain necessary at deployment-grade quality. In our evaluation, F1 serves as a *control variable* verifying that constraint enforcement does not degrade extraction quality, not as an optimization target. The contribution of this work is the constraint enforcement method itself; the zero-shot setting stress-tests constraint robustness under maximum model uncertainty, while composability with fine-tuning (§5.1) validates practical deployment value.

Decoding-Time vs. Post-Hoc Enforcement. Post-hoc filtering (generating triples without constraints, then removing schema-violating predictions) yields higher precision and consequently higher F1 than OntoCoDec on both datasets (SciERC 8.63 vs. 8.36; DocRED 9.61 vs. 8.23). However, this precision gain comes at substantial cost. On DocRED, post-hoc filtering discards 48.1% of all predicted triples, including 127 true positives (11.2% of correct predictions); recall drops from 9.46% to 8.40%. Entity type violations remain unresolved (ET = 2.47% vs. OntoCoDec's 0%), as filtering targets only domain/range and vocabulary violations. OntoCoDec's decoding-time approach avoids this information loss by guiding generation toward compliant outputs rather than discarding non-compliant ones, preserving both recall and entity type validity. The two strategies are complementary: post-hoc filtering can serve as an additional safety net for residual violations that decoding-time enforcement does not fully resolve.

Cross-Field Violation Distribution. Table 9 reports the complete distribution of conflicting-type pairs for SciERC S1 (245 CF violations, all pairwise; no mention triggered ≥3 types). Conflicts involving the GENERIC type account for 63.3% (155/245) of all violations. GENERIC is SciERC's anaphoric/placeholder label ("this approach", "the method"), which competes with every content type depending on whether the mention is used anaphorically or referentially, a distinction the per-slot constrained decoder cannot resolve without document-level coreference.

The pattern generalizes across models and datasets: on Llama-3.1-8B SciERC, GENERIC↔OTHERSCIENTIFICTERM remains the top pair (53/162, 32.7%); on DocRED, the analogous catch-all type MISC drives 76.6% of Qwen's CF violations. Under unconstrained decoding, CF rates collapse to ≤1% on both datasets; CF violations are a side-effect of per-slot type commitment rather than a model deficiency.

Per-Relation DR Elimination under Fine-Tuning. Table 10 presents the per-relation breakdown of DR elimination when OntoCoDec is composed with LoRA fine-tuning on SciERC. Fine-tuning alone eliminates DR on four of seven relations but leaves non-zero violations on the three tightest-constraint relations: EVALUATE-FOR

Relation	κ	LoRA Unconstr. DR (%)	LoRA Unconstr. n	LoRA + S1 DR (%)	LoRA + S1 n
Evaluate-for	0.361	13.13	99	**0**	98
Part-of	0.417	5.77	52	**0**	52
Compare	0.528	5.00	40	**0**	40
Conjunction	0.306	1.31	153	**0**	153
Feature-of	0.278	1.35	74	**0**	74
Used-for	0.028	0.16	634	**0**	635
Hyponym-of	0.417	0	—	**0**	—

Table 10: Per-relation DR elimination on SciERC with LoRA fine-tuning (Qwen3-8B). κ: constraint tightness; n: predicted triple count. Used-for gains one triple from Evaluate-for redirection.

Dataset	Model	Mode	DR↓	RV↓	ET↓	CF	PF	E-F1↑	R-F1↑
SciERC	Qwen3-8B	Unconstrained	1.95	0.00	0.00	0.00	0.00	67.05	43.86
		Grammar	1.90	0.00	0.00	0.00	0.00	67.05	43.86
		Semantic	0.60	0.00	0.00	0.18	0.00	67.05	43.69
		S1	**0**	0.00	0.00	1.18	0.00	67.05	43.77
	Llama-3.1-8B	Unconstrained	1.11	0.00	0.00	0.00	0.00	67.76	45.45
		Grammar	1.11	0.00	0.00	0.00	0.00	67.76	45.45
		Semantic	0.65	0.00	0.00	0.10	0.00	67.76	45.47
		S1	**0.09**	0.00	0.00	0.46	0.00	67.76	45.47
DocRED	Qwen3-8B	Unconstrained	0.56	0.27	0.00	0.08	7.11	82.35	40.89
		Grammar	0.61	0.00	0.00	0.11	7.01	82.19	40.85
		Semantic	0.21	0.23	0.00	0.27	7.52	82.15	40.89
		S1	**0.00**	0.00	0.00	0.22	8.52	81.87	40.79
	Llama-3.1-8B	Unconstrained	0.49	0.15	0.00	0.08	2.30	84.77	44.55
		Grammar	0.51	0.00	0.00	0.12	2.30	84.04	43.77
		Semantic	0.30	0.15	0.00	0.16	2.40	84.02	43.90
		S1	**0.00**	0.00	0.00	0.26	2.51	84.71	44.70

Table 11: Four-way constraint ablation on fine-tuned models (LoRA, 2 models × 2 datasets). DR/RV/ET/CF: violation rates (%); PF: parse failure rate (%); F1 ×100. Grammar layer is a no-op post-fine-tuning (DR change <0.2 pp in all 4 settings; RV and ET already 0% without constraints). Semantic layer is the sole effective DR defense, providing 39–69% reduction. F1 cost is negligible: max ΔR-F1 = −0.10 pp (Qwen DocRED, full vs. unconstrained).

(κ=0.361, 13.13%), PART-OF (κ=0.417, 5.77%), and COMPARE (κ=0.528, 5.00%). Adding OntoCoDec constraints eliminates all remaining violations across all relations. One EVALUATE-FOR triple (99 → 98) is redirected to USED-FOR (634 → 635) by token-level masking: the head entity's registered type falls outside EVALUATE-FOR's domain, and the semantic mask steers generation to the nearest valid relation.

Fine-Tuned Four-Way Ablation. Table 11 presents the four-way constraint ablation on fine-tuned models across both datasets, isolating each layer's contribution when the base model has already internalized schema knowledge.

The grammar layer is effectively a no-op for fine-tuned models across all four settings: RV and ET are already 0% without constraints, and grammar-only DR change is <0.2 pp (4/4 settings).

Mode	Violation Rates (%) DR↓	ΔDR	RV↓	ET↓	CF	Tot.↓	F1 Rel.↑	Ent.↑
Unconstrained	18.46	—	29.66	2.48	0.10	18.56	7.74	58.15
Grammar	26.17	↑42%	**0.03**	**0**	0.07	26.24	**8.31**	**59.38**
Semantic	13.84	↓25%	29.87	4.22	1.05	14.89	7.70	57.81
S1	**10.21**	**↓45%**	**0.03**	**0**	2.57	**12.78**	8.23	59.37

Table 12: Four-way ablation on DocRED (998 documents, Qwen3-8B, zero-shot). ΔDR: relative change vs. unconstrained. Tot. = DR + CF (total type-violation burden). Best per column in **bold** (CF excluded; Tot. included). F1 ×100. Bootstrap significance: R-F1 +0.50 pp ($p < 0.001$). Grammar-only ↑DR: vocabulary enforcement makes OOV outputs evaluable, exposing latent violations (see §5.2).

The semantic layer is the sole effective DR defense under fine-tuning, independently reducing DR by 39–69% across all settings (SciERC: Qwen −69%, Llama −41%; DocRED: Qwen −63%, Llama −39%). Adding both layers achieves near-complete elimination (DR ≤ 0.09% in all 4 settings): even after fine-tuning, the semantic layer's pair-based enforcement captures violations that training alone cannot eliminate, specifically on tight-constraint relations where the valid type-pair space is small. F1 cost is negligible: the maximum R-F1 decrease across S1 vs. unconstrained is −0.10 pp (Qwen DocRED).

DocRED Zero-Shot Ablation. Table 12 presents the four-way constraint ablation on DocRED under zero-shot decoding, parallel to the SciERC ablation in Table 1.

Residual DR Error Analysis. Across all fine-tuned settings (2 models × 2 datasets × S1), only 36 domain/range violations remain. Manual inspection reveals four root causes:

1. *Entity type errors* (39%): the LLM assigns an incorrect entity type (e.g., labeling a geographic entity as LOC when the gold standard requires MISC), causing the semantic mask to permit a domain/range-incompatible relation.
2. *Schema coverage gaps* (33%): the frequency-based constraint table, constructed from training-set co-occurrence statistics, does not cover long-tail type pairs that are ontologically valid but rare. These violations are irreducible by decoding-time constraints alone.
3. *Incorrect relation selection* (17%): the model selects a semantically inappropriate relation whose domain/range constraints are incompatible with the correctly typed entities.
4. *Direction reversal* (11%): the model assigns the

Setting	Model	Dataset	Reject%	R-F1$_{ph}$	R-F1$_{cd}$	Winner
ZS	Qwen	SciERC	16.8	8.63	8.36	ph
	Llama	SciERC	13.9	2.71	2.54	ph
	Mistral	SciERC	20.3	3.94	3.67	ph
	Qwen	DocRED	48.1	9.61	8.23	ph
	Llama	DocRED	42.2	7.44	6.27	ph
	Mistral	DocRED	**73.7**	3.85	**5.52**	cd
FT	Qwen	SciERC	1.95	43.94	43.77	ph
	Llama	SciERC	1.11	45.23	45.47	cd
	Qwen	DocRED	1.03	41.11	40.79	ph
	Llama	DocRED	0.64	43.92	44.70	cd

Table 13: Post-hoc filtering (ph) vs. constrained decoding (cd). In zero-shot settings, post-hoc F1 is higher in 5/6 groups (precision-dominated), but constrained decoding preserves higher recall in all 6/6 groups. At reject rate >70% (Mistral DocRED), post-hoc F1 collapses below constrained. Under fine-tuning, rejection rates drop to 0.6–2.0% and the two strategies are equivalent ($\Delta \leq 0.55$ pp; each wins two groups). F1 $\times 100$.

correct relation but inverts head/tail roles, placing entities in positions that violate the relation's asymmetric domain/range specification.

Two dataset-specific patterns emerge. Outside Llama DocRED, all residual violations (3 of 36, spanning Llama SciERC and Qwen DocRED) trace exclusively to schema coverage gaps: the model's entity typing is accurate, and the constraint system has reached its theoretical ceiling given the frequency-based constraint table. On DocRED, Llama-3.1-8B contributes 33 of 36 residual violations (92%), with entity type errors as the dominant cause (14/33, 42%). This asymmetry reflects Llama's weaker fine-tuned entity typing on DocRED's larger ontology (96 relations, 7 entity types) compared to Qwen, which achieves near-zero residual. The implication is twofold: (i) the irreducible floor from schema gaps is small (∼3 violations across 4 settings), so the constraint system operates near its theoretical optimum; (ii) residual violations from entity typing errors are addressable through stronger base models or improved entity typing modules, without modifying the constraint architecture.

Post-Hoc Filtering: Extended Comparison. Table 13 presents the full comparison across all six zero-shot and four fine-tuned configurations.

The crossover point occurs when post-hoc rejection exceeds ∼70%: Mistral DocRED rejects 73.7% of predicted triples, discarding enough true positives that constrained decoding's recall advantage dominates (+1.67 pp). OntoCoDec's advantage thus scales with violation severity: precisely when constraints are most needed, post-hoc filtering is most destructive. Under fine-tuning, both strategies converge because few violations remain to filter or constrain ($\Delta \leq 0.55$ pp across all four FT groups, with each strategy winning two).

Zero-Shot Runtime Profiling. On SciERC (Qwen3-8B, zero-shot greedy, NVIDIA RTX PRO 6000 Blackwell Server Edition), unconstrained inference averages 4.163 s/doc and S1 constrained decoding averages 4.710 s/doc (+13.1% overhead). On DocRED (same configuration), relative overhead is lower due to amortization: the fixed per-token constraint cost is spread over longer generation sequences. XGrammar automaton compilation takes 0.044 s; the full constraint pipeline (schema parsing, JSON Schema generation, grammar compilation) takes 0.33 s on SciERC (Table 41), amortized to zero over the evaluation set.

Fine-Tuned Latency Profiling. Under LoRA fine-tuning (same hardware and dataset), unconstrained inference averages 2.612 s/doc and S1 constrained decoding averages 2.873 s/doc (+10.0% overhead, n=551). Fine-tuned models generate shorter, more structured outputs than zero-shot models (2.612 s vs. 4.163 s/doc unconstrained); absolute generation time drops by 37%. The relative overhead (+10.0%) is slightly lower than zero-shot (+13.1%), consistent with shorter sequences requiring fewer constraint checks per document.

Relation Matching Protocol. Relation F1 evaluation uses fuzzy normalization to handle model-specific formatting variants (e.g., underscores vs. spaces, abbreviations). A predicted relation r_p matches gold relation r_g if: (1) exact string match after case-insensitive normalization, or (2) token overlap >80% *and* edit distance ≤ 2. This protocol affects primarily Mistral, which produces underscore-separated variants (e.g., "head_of_government" for "head of government"). On Mistral DocRED, fuzzy matching contributes 98 additional TPs (+0.84 pp R-F1; strict R-F1 = 3.50% vs. fuzzy 4.34%). For Qwen and Llama under grammar constraints, the grammar layer enforces exact relation names, making fuzzy matching irrelevant (0 additional matches). All F1 figures in the main text use fuzzy normalization consistently across all models and conditions.

Per-Relation Violation Landscape. Table 14 compares violation rates under unconstrained and

Table 14: Violation rates (%) under unconstrained and schema-informed prompting (Qwen3-8B, zero-shot).

Dataset	Prompt	DR↓	RV↓	ET↓	CF↓
SciERC	Unconstrained	15.4	1.4	0.4	0.1
	Schema-informed	5.8	0.3	**0.0**	—
DocRED	Unconstrained	18.5	29.7	2.5	0.1
	Schema-informed	12.8	18.4	0.2	0.2

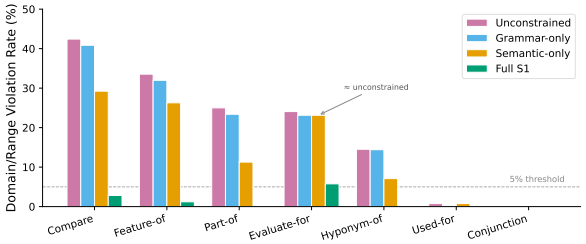


Figure 3: Per-relation DR violation rates on SciERC (Qwen3-8B, unconstrained). Rates correlate with constraint tightness $\kappa(r)$. Llama-3.1-8B shows similar per-relation patterns (Appendix D).

schema-informed prompting. Figure 3 shows the per-relation DR distribution on SciERC.

WikiEvents Dataset Details. WikiEvents (Li et al., 2021) is a document-level event argument extraction benchmark grounded in the DARPA KAIROS ontology. It defines 56 event types, 80 argument roles, and 27 entity types, with 245 (event_type, role)→allowed_entity_types constraints. The grammar layer targets EV (event vocabulary), RoV (role vocabulary), and ET; the semantic layer enforces AC (argument constraint) via role-conditioned masking, analogous to RE domain/range enforcement.

Constraint Tightness in WikiEvents. WikiEvents role-level tightness (with $|\mathcal{E}|$=27) has mean $\bar{\kappa}$=0.82 and median 0.89; 84% of roles have $\kappa \geq 0.80$ (4 or fewer valid entity types out of 27). The tightness-violation pattern extends to event extraction: roles with $\kappa \geq 0.90$ exhibit the highest AC rate (50%), compared to 21% for medium-tightness ($0.80 \leq \kappa < 0.90$) and 30% for loose ($\kappa < 0.80$) roles; this is consistent with the RE observation that constraint tightness correlates with violation frequency. However, the correlation is modulated by entity type prototypicality: semantically intuitive roles (e.g., Transportation.Vehicle→VEH) show 0% AC despite κ=0.963, while roles that require rare KAIROS-specific types (e.g., Crime→CRM, Sentence→SEN) show near-100% AC because LLMs default to common types (PER, ORG). The loose group's elevated rate (30%) is driven by high-frequency roles such as Attack.Target (126 args, 34% AC, κ=0.59), where systematic model errors outweigh the permissive constraint space.

Lucky Violation Analysis. Table 15 quantifies the recall cost of DR elimination. A "lucky violation" is a constraint-removed triple that happens to match a gold label—an ontologically invalid prediction that is nonetheless correct. Across four fine-tuned configurations, lucky violations average 0.49% of gold triples (range 0.01–0.51%), confirming that nearly all removed DR violations are false positives. On Llama SciERC, all five lucky violations are recovered as legal type paths under constrained decoding (zero churn): the model finds an ontology-compliant route to the same prediction.

Model	Dataset	DR (%) unc→con	Lucky (% gold)	ΔRecall (pp)	Churn (%)
Qwen3-8B	SciERC	1.95→0	4 (0.41%)	−0.10	0.22
Llama-8B	SciERC	1.11→0.09	5 (0.51%)	±0.00	0
Qwen3-8B	DocRED	1.03→0.01	1 (0.01%)	−0.84	2.05
Llama-8B	DocRED	0.64→0.21	1 (0.01%)	−0.10	0.86
Average			*0.49%*	−0.26	0.78

Table 15: Lucky violation analysis across four LoRA fine-tuned configurations. Lucky: count (and % of gold triples) of constraint-removed violations matching gold labels. ΔRecall: recall change from unconstrained to constrained. Churn: fraction of gold-matching triples whose predictions change (including recovery via legal type paths).

Trie-Only Baseline (GCD-Equivalent). Table 16 isolates the effect of vocabulary membership constraints without JSON structural enforcement. The trie constrains entity type and relation tokens to schema-valid values (achieving ET=0%, RV=0% across all configurations), functionally equivalent to GCD's (Geng et al., 2023) constrained decoding. However, without grammar-level JSON enforcement, models systematically omit entity type fields (83% of triples) or produce malformed JSON (0–51% parse failure), causing DR to rise to 15–96%—far worse than unconstrained baselines. The effect is most dramatic for Llama-8B LoRA: 50.8% parse failure causes R-F1 to collapse from 45.47 (OntoCoDec S1) to 19.95, demonstrating that the grammar layer is not merely helpful but a structural prerequisite—vocabulary constraints require JSON scaffolding to reach the positions where they apply.

Model	Setting	ET ↓	RV ↓	DR ↓	CF	R-F1 ↑	PF (%)
Qwen3-8B	LoRA	0	0	95.5	0.09	41.16	9.1
	ZS	0	0	29.3	0	8.64	0
Llama-8B	LoRA	0	0	15.2†	0	19.95	50.8
	ZS	0	0	84.4	0.36	2.90	0

Comparison with OntoCoDec S1:

		S1 DR (%)	Trie DR (%)	DR Gap (pp)
Qwen3-8B	LoRA	0	95.5	95.5
Qwen3-8B	ZS	0.93	29.3	28.4
Llama-8B	LoRA	0.09	15.2	15.1
Llama-8B	ZS	0.08	84.4	84.3

Table 16: Trie-only baseline (GCD-equivalent, vocabulary membership without JSON grammar) on SciERC. Top: full trie-only metrics (ET/RV/DR/CF: violation rates %; R-F1 ×100; PF: parse failure rate). Trie achieves ET=0%, RV=0% but DR rises to 15–96% because models omit type fields without structural enforcement. Bottom: DR gap between OntoCoDec S1 and trie-only, quantifying the grammar layer's contribution. †50.8% parse failure rate.

Comparison with Grammar-Based Constrained Decoding. Table 17 summarizes the relationship between OntoCoDec's constraint layers and existing grammar-based constrained decoding methods.

Method	Constraint Type	Level	DR Red.	CF	State
GCD (Geng et al., 2023)	CFG (JSON schema)	Grammar	∼3%	N/A	Stateless
BoostCD (Sakota and West, 2025)	CFG + beam rerank	Grammar	∼3%†	N/A	Stateless
Trie-Only (Tab. 16)	Prefix trie	Vocab	0%‡	N/A	Stateless
vLLM `guided_json` (Kwon et al., 2023)	JSON Schema	Grammar	∼3%§	N/A	Stateless
OntoCoDec Grammar	CFG (XGrammar)	Grammar	∼3%	N/A	Stateless
OntoCoDec S1	CFG + type constr.	Gram.+Sem.	94–100% (FT)	≤1.4% (FT)	Entity reg.

Table 17: Comparison with grammar-based constrained decoding methods. †Equivalent mechanism; BoostCD adds beam-level reranking but the CFG enforcement is identical. ‡Vocabulary-only constraints without JSON structure cause DR to *increase* (15–96%; Table 16). §Experimentally verified: DR=15.26% vs. grammar-only 14.84% (Δ<0.5 pp; Table 18).

Our grammar layer implements the same CFG-based token masking as GCD and BoostCD—the mechanisms are formally identical (JSON schema → pushdown automaton → per-step allowed token set). Table 1 row 2 therefore constitutes a direct comparison: grammar-only enforcement achieves RV/ET→0% but only ∼3% DR reduction, identical to what GCD-family methods would achieve on IE tasks.

External verification via vLLM. To confirm this equivalence experimentally, we evaluated vLLM's `guided_json` mode (Kwon et al., 2023)—an independent CFG enforcement implementation—on SciERC under identical conditions (Qwen3-8B, zero-shot, 551 documents).

All metrics differ by <0.5 pp (Table 18), confirming that any CFG-based constrained decoding tool—whether GCD, BoostCD, or vLLM—enforces the same constraint class and leaves the same ∼15% DR gap. The critical gap these methods cannot address is cross-field type consistency (Level 2), which requires the stateful semantic layer that maintains an entity registry across relation slots and enforces per-relation domain/range constraints at each type-bearing position.

Method	DR ↓	RV ↓	ET ↓	CF	R-F1 ↑
Grammar-Only (XGrammar)	14.84	0	0	9.90	8.23
vLLM `guided_json`	15.26	0	0	—	8.75
Δ	+0.42	0	0	—	+0.52

Table 18: GCD-equivalence verification: OntoCoDec grammar-only vs. vLLM `guided_json` on SciERC (Qwen3-8B, zero-shot, 551 documents). All differences <0.5 pp, confirming that CFG-based approaches—regardless of implementation—leave ∼15% DR violations unaddressed.

70B-class fine-tuned ablation. Table 19 presents the constraint ablation on Llama-3.1-70B-Instruct (AWQ INT4) with LoRA fine-tuning on SciERC (551 documents).

Mode	DR ↓	RV ↓	ET ↓	CF	R-F1 ↑	E-F1 ↑
Unconstrained	1.50	0	0	0	44.73	64.86
Grammar	1.50	0	0	0	44.63	64.82
Semantic	0.60	0	0	9	44.73	64.86
S1	**0**	0	0	15	44.53	64.76
S2	**0**	0	0	18	36.28	62.35
Beam4 S1	**0**	0	0	**1**	**44.81**	**64.89**

Table 19: 70B-class (Llama-3.1-70B-Instruct (AWQ INT4)) LoRA ablation on SciERC (551 docs). CF: violation count. F1 ×100.

The grammar layer provides no independent DR benefit at 70B (1.50% → 1.50% on SciERC; on DocRED zero-shot, DR actually increases +3.58 pp due to RV→DR conversion in the high-$\bar{\kappa}$ schema; Table 21). The semantic layer is the sole effective defense (1.50% → 0.60%, 60% reduction) with no R-F1 cost (Δ=0 pp), and full S1 eliminates all DR. S2 eliminates DR but causes severe R-F1 degradation (−8.45 pp), indicating that optimal constraint granularity is scale-dependent: the additional S2 constraints become overly restrictive at 70B, where stronger baseline generation makes fine-grained intervention counterproductive. Beam (k=4) S1 remains the optimal configuration (DR=0%, CF=1, R-F1=44.81). Notably, S2 increases CF from 15 (S1) to 18 despite its cross-field consistency design, suggesting that over-constraining induces cascad-

ing generation degradation that creates new violations.

70B-class DocRED results. Table 20 presents the five-way constraint ablation on Llama-3.1-70B-Instruct (AWQ INT4) with LoRA fine-tuning on DocRED (998 documents).

Mode	DR↓	RV↓	ET↓	CF	R-F1↑	E-F1↑
Unconstrained	0.22	0.57	0	2	43.80	80.34
Grammar	0.38	0	0	0.06%†	43.96	80.13
Semantic	**0**	0.51	0	0.21%†	43.61	80.06
S1	**0**	0	0	0.18%†	43.95	79.38
Beam4 S1	**0**	0	0	**0**	**48.14**	**85.57**

Table 20: 70B-class (Llama-3.1-70B-Instruct (AWQ INT4)) LoRA ablation on DocRED (998 docs). CF: violation count except †reported as percentage (raw count unavailable). F1 ×100. PF: greedy S1 12.5% (25/200 docs), beam4 S1 0.5% (1/200).

The DocRED LoRA ablation mirrors the SciERC pattern: grammar eliminates RV (0.57% → 0%) but DR increases slightly (+0.16 pp); semantic-only independently eliminates DR (0.22% → 0%) but retains RV (0.51%). The complementary layer interaction is stable across datasets under LoRA fine-tuning. The +4.19 pp R-F1 gain from beam search (43.95 → 48.14) is largely attributable to parse failure recovery: greedy S1 fails to parse 12.5% of documents (25/200), while beam reduces this to 0.5% (1/200), recovering ∼24 documents that contribute no triples under greedy decoding. The controlled comparison on SciERC—where both greedy and beam achieve 0% parse failure—isolates the pure beam quality gain at +0.28 pp, consistent with the 8B pattern (Appendix F).

70B-class DocRED zero-shot ablation. Table 21 presents the layer ablation on DocRED under zero-shot decoding at 70B.

Mode	DR↓	RV↓	ET↓	CF	R-F1↑	E-F1↑
Unconstrained	13.45	7.88	0	0	13.25	70.14
Grammar	17.03	0	0	0	13.23	70.12
Semantic	1.66	8.24	0.12	132	13.05	70.10
S1	**0**	0	0	173	13.25	70.05

Table 21: 70B-class zero-shot layer ablation on DocRED. CF: violation count. F1 ×100.

Grammar-only eliminates RV (7.88% → 0%) but DR increases by +3.58 pp (13.45% → 17.03%): forced vocabulary mapping in DocRED's high-$\bar{\kappa}$ schema ($\bar{\kappa}$=0.842, ∼16% type pairs valid) converts RV into DR violations. Semantic-only independently achieves 87.7% DR reduction (13.45% → 1.66%) with no R-F1 cost, confirming the pattern from SciERC (84%; Table 19). Full S1 eliminates all DR (13.45% → 0%) with zero R-F1 cost (13.25 = 13.25), confirming that the two layers compose correctly despite their opposing individual DR effects. CF increases from 0 (unconstrained) to 173 (S1), consistent with the pattern that stricter constraint enforcement in ZS introduces more cross-field inconsistencies. The two layers serve complementary roles: grammar provides the RV elimination foundation that semantic-layer DR enforcement requires, while the semantic layer addresses the DR that grammar alone cannot—and in high-$\bar{\kappa}$ schemas, exacerbates.

D Per-Relation Violation Analysis (Llama-3.1-8B)

Table 22 presents the per-relation domain–range violation breakdown for Llama-3.1-8B on DocRED under S1 constraints. The Llama baseline exhibits an RV rate of 16.03%, substantially lower than Qwen's 29.66%, which indicates stronger baseline vocabulary compliance. S1 completely eliminates RV violations and reduces the residual DR rate to 2.55% (418/16,368 triples), with only three relations exceeding 10%.

Table 22: Top 10 per-relation DR violation rates for Llama-3.1-8B on DocRED under S1 constraints. Relations with >20 triples and 0% DR (e.g., P131, P206, P361, P170, P86) are omitted.

Relation	PID	Triples	DR Violations	DR Rate (%)
dissolved/demolished	P576	73	24	32.9
followed by	P156	117	24	20.5
unemployment rate	P1198	15	3	20.0
end time	P582	130	21	16.2
instance of	P31	290	37	12.8
product/material produced	P1056	98	11	11.2
location of formation	P740	81	8	9.9
head of state	P35	41	3	7.3
applies to jurisdiction	P1001	637	45	7.1
conflict	P607	453	30	6.6

Compared with Qwen, which retains a 10.21% residual DR under S1 concentrated in P1198 (83% fallback) and P171 (78% fallback), Llama's residual DR on the same relations is markedly lower (P1198 reaches only 20.0%) and no relation-level fallback concentration comparable to Qwen is observed. The substantially lower residual DR for Llama (2.55% vs. Qwen's 10.21%) on the same ontology confirms that constraint tightness alone does not determine enforcement effectiveness: the

model's baseline vocabulary compliance (RV) modulates the fraction of triples eligible for semantic-layer enforcement.

E Sampling Robustness

Table 23: Constraint robustness under nucleus sampling (T=0.7, p=0.9) vs. greedy decoding (Qwen3-8B, S1 constraints).

Dataset	Decoding	ET%	RV%	DR%	CF%	E-F1	R-F1
SciERC (551)	Greedy	0	0	0.93	9.90	30.98	8.36
	Sampling	0	0	0.75	9.90	31.28	7.42
DocRED (114†)	Greedy	0	0.03	10.21	2.57	59.37	8.23
	Sampling	0	0	10.31	12.63	57.05	7.92

†Subset of 114 documents; full DocRED sampling requires ∼300h due to extended generation under high-temperature sampling with complex schemas.

F Beam Search Compatibility

OntoCoDec's constraint enforcement is designed to be decoding-strategy agnostic. The main experiments (§5) use greedy decoding; Appendix E validates nucleus sampling. Here we verify compatibility with beam search, completing coverage of the three standard decoding strategies.

Implementation challenges. Two stateful components require adaptation for beam search. (1) The grammar layer (XGrammar) maintains an internal `GrammarMatcher` state that tracks the generated token sequence; beam reordering invalidates this state because beams are permuted across steps. Our solution instantiates a fresh `GrammarMatcher` per beam per step and replays the beam's current token sequence through XGrammar's stateless accept API (64 lines of code). (2) The semantic layer's `TypeConstraintProcessor` accumulates entity and relation registries that assume a single generation path. Our solution isolates state per beam by calling `detect()` on the full partial output at each step, reconstructing all entity/relation state from scratch for each beam independently (61 lines of code). Both adaptations add ∼125 lines total without modifying the existing greedy/sampling logic.

Results. Table 24 compares beam search (k=4, `early_stopping=True`) against greedy decoding under S1 constraints across six model-dataset combinations.

Table 24: Beam search (k=4) vs. greedy decoding under S1 constraints (zero-shot). SciERC: 551 documents; DocRED: matched 50-document subsets (same documents for greedy and beam); full 998-doc DocRED results in Table 27. ET = entity type, RV = relation vocabulary, DR = domain/range, CF = cross-field consistency violation rates (%); F1 × 100.

Model / Dataset	Decoding	ET	RV	DR↓	CF↓	E-F1↑	R-F1↑
Qwen3-8B / SciERC (551)	Greedy	0	0	0.93	9.90	30.98	8.36
	Beam (k=4)	0	0	**0**	**0.53**	32.00	8.56
Qwen3-8B / DocRED (50)	Greedy	0	0	8.69	2.08	59.08	8.33
	Beam (k=4)	0	0	**0**	**0.55**	58.64	9.37
Llama-3.1-8B / SciERC (551)	Greedy	0	0	0.08	6.56	23.49	2.54
	Beam (k=4)	0	0	**0**	**0.93**	25.04	3.33
Llama-3.1-8B / DocRED (50)	Greedy	0	0	1.58	6.85	55.35	6.12
	Beam (k=4)	0	0	**0**	**0.94**	55.62	9.47
Mistral-7B / SciERC (551)	Greedy	0	0	13.40	1.27	27.46	3.67
	Beam (k=4)	0	0	13.79	**0.57**	26.66	4.37
Mistral-7B / DocRED (50)	Greedy	0	0	28.37	0.34	51.60	4.19
	Beam (k=4)	0	0	**21.37**	**0.19**	51.78	5.70

Discussion. For calibrated models (Qwen3-8B, Llama-3.1-8B), beam search achieves DR=0% across all four model-dataset combinations (both SciERC with 7 entity types, 551 documents, and DocRED with 96 relation types, 50 documents), regardless of schema complexity. On SciERC, beam search eliminates DR for both Qwen (0.93% → 0%) and Llama (0.08% → 0%), and substantially reduces CF (Qwen: 9.90% → 0.53%; Llama: 6.56% → 0.93%). On DocRED (matched 50-document subsets), beam search eliminates DR for both Qwen (8.69% → 0%) and Llama (1.58% → 0%), with CF reductions of 74% and 86% respectively. Llama DocRED shows the largest R-F1 gain (+3.4 pp), consistent with beam search recovering valid type assignments that greedy decoding missed due to constraint fallback. The CF reduction is an indirect effect: beam search reduces constraint-fallback frequency by exploring alternative token sequences that satisfy both grammar and semantic masks simultaneously, avoiding the fallback-to-grammar path that produces cross-field inconsistencies under greedy decoding.

Mistral-7B-v0.3 constitutes a boundary condition on both datasets. On SciERC (7 entity types), beam search does not reduce DR (13.4% → 13.8%), though CF drops by 55% relative (1.27% → 0.57%) and R-F1 improves by +0.7 pp; the unchanged DR confirms that all residual violations stem from entity type misclassification, not constraint-search failure. On DocRED (96 relation types), beam search reduces DR by 25% relative (28.4% → 21.4%): the more complex schema introduces constraint-search failures that beam exploration resolves, but the entity

misclassification component remains. This contrast reveals that beam search resolves constraint-search failure (where greedy decoding encounters $G(t) \cap S(t) = \emptyset$ conflicts) but cannot overcome fundamental model calibration limitations in entity typing.

Beam width ablation. Table 25 varies beam width $k \in \{1, 2, 4, 8\}$ on Qwen3-8B / SciERC (551 documents) under S1 constraints.

Table 25: Effect of beam width on constraint satisfaction and extraction quality (Qwen3-8B, SciERC, 551 documents, S1 constraints). k=1 corresponds to greedy decoding.

k	DR ↓	CF ↓	E-F1 ↑	R-F1 ↑
1 (greedy)	0.93	9.90	30.98	8.36
2	0.35	4.12	31.54	**8.75**
4	**0**	0.53	32.00	8.56
8	**0**	**0.08**	**32.18**	8.39

DR elimination at $k \geq 4$ confirms that constraint-search failure is resolved with moderate exploration width. CF continues to decrease with larger beam (0.08% at k=8, only 2 residual violations), while entity extraction quality improves monotonically. k=4 represents the optimal cost-effectiveness trade-off: it eliminates all DR violations at roughly half the compute cost of k=8.

Beam width–latency Pareto analysis (LoRA). Figure 4 and Table 26 extend the beam width analysis to LoRA fine-tuning, where the constrained–unconstrained interaction shifts qualitatively: LoRA + S1 already achieves DR=0% at k=1 (greedy), so beam search targets CF reduction rather than DR elimination (contrast: zero-shot requires $k \geq 4$ for DR=0%). Among constrained configurations, k=2 is Pareto optimal (R-F1 44.60, CF 0.11%, 1.39× latency); $k \geq 4$ yields diminishing F1 returns with increasing latency (1.63–2.03×).

Composability with fine-tuning. Under greedy decoding, the fine-tuned Qwen3-8B (LoRA, SciERC) with S1 constraints already achieves DR=0% but retains CF=1.18% (21 cross-field violations across 551 documents), with E-F1=67.05 and R-F1=43.77. Adding beam search (k=4) eliminates all residual CF violations (1.18% → 0%), yielding zero violations across all four metrics (ET=0%, RV=0%, DR=0%, CF=0%) while preserving extraction quality (E-F1=66.88, R-F1=43.60; changes within ±0.5 pp). Llama-3.1-8B (LoRA, SciERC) with S1 constraints and beam search (k=4) likewise achieves zero violations across all four metrics (ET=0%, RV=0%, DR=0%, CF=0%) with R-F1=46.18; the zero-violation result thus generalizes across calibrated model families. The three constraint mechanisms (grammar masking, semantic domain-range filtering, and beam exploration) are thus fully composable: each addresses a distinct violation source without interfering with the others or degrading task performance.

Table 26: Beam width sweep on SciERC (Qwen3-8B LoRA, 551 documents). UNC: unconstrained; CON: schema-constrained. Latency relative to k=1 unconstrained. Highlighted row: Pareto-optimal configuration.

k	Mode	E-F1	R-F1	DR (%)	CF (%)	Latency (s)	Rel. Lat.
1	UNC	66.88	42.95	2.11	0	2.494	1.00×
	CON	66.88	42.95	**0**	0.50	2.932	1.18×
2	UNC	67.79	44.75	1.54	0	2.954	1.18×
	CON	67.71	**44.60**	**0**	0.11	3.463	1.39×
4	UNC	67.49	43.72	1.89	0	3.485	1.40×
	CON	67.22	43.60	**0**	0	4.069	1.63×
8	UNC	66.40	43.17	1.77	0	4.194	1.68×
	CON	66.36	43.65	**0**	0	5.058	2.03×

The initial 50-document DocRED subset (Table 24) was subsequently validated on the full 998-document set (Table 27): both models achieve DR=0%, confirming subset representativeness. Llama shows the larger improvement (CF 8.54% → 1.00%, 88% reduction; R-F1 +2.6 pp), consistent with beam search resolving constraint-search failures that greedy decoding cannot.

Table 27: Zero-shot beam search (k=4) under S1 constraints on the full DocRED dev set (998 documents). ET/RV/DR/CF: violation rates (%); PF: parse failure rate (%); F1 ×100.

Model	ET	RV	DR ↓	CF ↓	E-F1 ↑	R-F1 ↑
Qwen3-8B	0	0	**0**	4.94	58.19	8.72
Llama-3.1-8B	0	0	**0**	1.00	57.81	8.83

Table 28 reports LoRA + beam-4 results on the full DocRED dev set (998 documents), extending the fine-tuning composability evaluation from SciERC to the higher-$\bar{\kappa}$ regime. Parse failure rates under LoRA beam-4 range from 1.3% to 2.9%, lower than the zero-shot rate.

Llama-3.1-8B achieves zero violations across all four metrics (ET, RV, DR, CF = 0%) on DocRED with LoRA + beam-4, extending the zero-violation result from SciERC to the high-$\bar{\kappa}$ regime. Qwen3-8B achieves DR=0% with near-zero CF (0.01%);

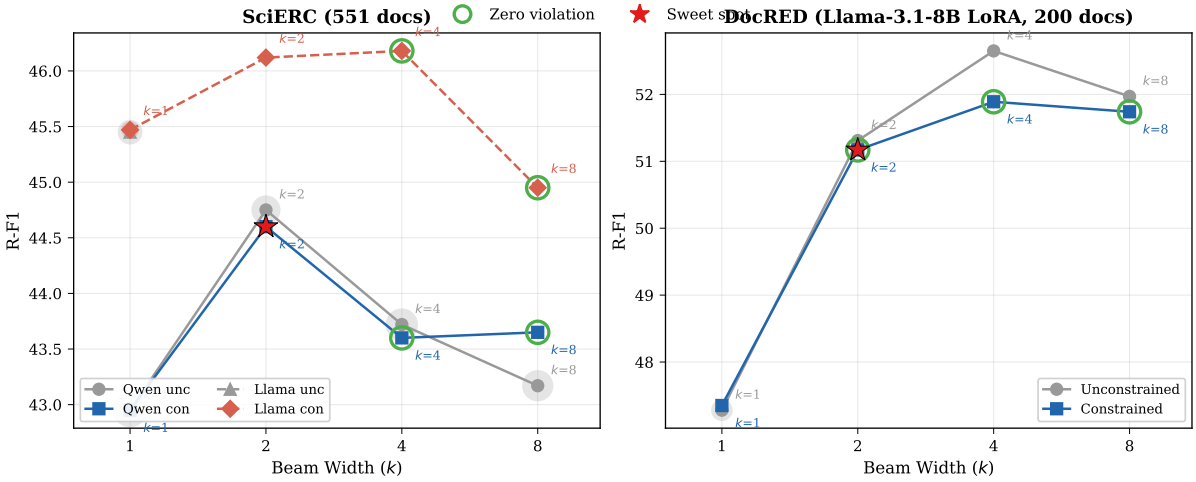


Figure 4: Beam width vs. R-F1 under LoRA fine-tuning. Left: SciERC (Qwen3-8B solid, Llama-3.1-8B dashed; 551 docs). Right: DocRED (Llama-3.1-8B, 200 docs). Bubble area $\propto$ total violation rate (DR+CF+RV); green rings mark zero-violation configurations. Star: Pareto-optimal (k=2).

Table 28: LoRA greedy vs. beam search (k=4) on DocRED (998 dev documents) under S1 constraints. ET = entity type, RV = relation vocabulary, DR = domain/range, CF = cross-field violation rates (%); PF = parse failure rate (%); F1 $\times$ 100.

Model	Decoding	ET	RV	DR↓	CF↓	E-F1↑	R-F1↑	PF
Qwen3-8B	Greedy con	0	0	0	0.22	81.87	40.79	7.8
	Beam-4 con	0	**0**	**0**	0.01	84.00	44.46	2.9
Llama-3.1-8B	Greedy con	0	0	0	0.26	84.71	44.70	2.6
	Beam-4 con	0	0	0	**0**	**85.23**	**47.17**	**1.3**

R-F1 is preserved ($\pm$0.06 pp). Combined with the sampling robustness results (Appendix E), these findings confirm that OntoCoDec's constraint enforcement is compatible with greedy, sampling, and beam search decoding without modification to the core constraint logic.

Latency overhead. Table 29 reports wall-clock latency for greedy and beam search (k=4) under LoRA fine-tuning.

Condition	s/doc	Overhead
Greedy unconstrained	2.612	1.00×
Greedy constrained (S1)	2.873	1.10×
Beam-4 unconstrained	3.446	1.32×
Beam-4 constrained (S1)	4.013	1.54×

Table 29: Latency per document (SciERC, 551 docs, Qwen3-8B LoRA FT, RTX PRO 6000 Blackwell). Overhead relative to greedy unconstrained.

Beam search itself adds 32% latency (4 beams); constraint overhead increases from 10% (greedy) to 17% (beam) because each beam independently maintains grammar and semantic state. The total 1.54× overhead is acceptable for offline batch IE; `early_stopping` limits additional compute to partial-sequence exploration.

We also explored a type-consensus constraint that forces identical entity types across all relations within a document. However, CF violations occur precisely where the consensus type conflicts with domain-range constraints, creating a fundamental CF↔DR trade-off. With DR-aware enforcement (skipping consensus when it would violate domain-range), the layer becomes a no-op (100% skip rate on 551 SciERC documents and 50 DocRED documents). This confirms that beam search resolves the CF↔DR tension through a fundamentally different mechanism: token-level exploration of paths satisfying both constraints simultaneously.

TCR reranking under beam search. TCR reranking with beam search (k=4) showed no effect on either Qwen3-8B or Llama-3.1-8B under LoRA (0/551 SciERC candidates changed): fine-tuned beam candidates are already type-consistent (TCR score=1.0 for all candidates), confirming that beam search implicitly subsumes post-hoc type consensus under fine-tuning. This further strengthens the beam search deployment recommendation: beam not only eliminates all four violation types but also renders post-hoc CF resolution unnecessary.

G Reproducibility Details

Fine-tuning configuration. All four fine-tuned models (Qwen3-8B and Llama-3.1-8B-Instruct on

SciERC and DocRED) use identical LoRA hyperparameters: rank r=16, α=32, dropout 0.05, no bias, targeting all attention and MLP projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`). Training uses AdamW with learning rate 2×10^{-4}, cosine schedule, warmup ratio 0.1, effective batch size 16, bf16 precision, gradient checkpointing, and completion-only loss for 3 epochs. SciERC trains on 1,762 examples (260 dev) for 333 steps; DocRED trains on 3,053 examples (998 dev) for 573 steps. Maximum sequence length is 2,048 (SciERC) and 4,096 (DocRED). All runs use seed 42.

Hardware and software. Training runs on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (98 GB VRAM). SciERC fine-tuning takes approximately 11 minutes per model; DocRED takes approximately 1.5 hours. Software versions: PyTorch 2.10, Transformers 4.57, PEFT 0.19, TRL 1.1, Accelerate 1.13.

Inference time. Inference timing is reported in Appendix C (NVIDIA RTX PRO 6000 Blackwell Server Edition GPU, same hardware as training). S1 constrained decoding adds +13.1% wall-clock overhead on SciERC (Qwen3-8B, zero-shot greedy), lower on DocRED due to amortization over longer sequences; under LoRA fine-tuning, overhead is +10.0% on SciERC.

Greedy decoding determinism. All main results use greedy decoding (temperature = 0, argmax token selection). Argmax is deterministic: the random seed controls `torch.manual_seed` for sampling and dropout, neither of which is active during greedy inference. We verify this with three seeds (42, 123, 456) across two LoRA fine-tuned models on all 551 SciERC documents. Seeds 123 and 456, evaluated on the same GPU, produce byte-identical outputs for both Qwen3-8B and Llama-3.1-8B. The residual variance in Table 30 stems entirely from cross-hardware floating-point divergence: seed 42 was evaluated on a different server with a different CUDA driver version. Even under this conservative upper bound, σ(E-F1) $\leq$ 0.16 pp, σ(R-F1) $\leq$ 0.43 pp, and σ(DR%) $\leq$ 0.08 pp. The constrained decoding DR% reduction ($-$2.06 pp for Qwen, $-$1.01 pp for Llama) exceeds this variance by >12× (signal-to-noise ratio > 12), confirming that observed improvements reflect genuine method effects rather than run-to-run noise. We therefore report single-run results without error bars; bootstrap confidence intervals (10,000 iterations) are provided for F1 metrics.

Table 30: Cross-seed variance for LoRA fine-tuned models on SciERC (551 documents, greedy decoding, seeds 42/123/456). Values: mean ± population σ across three seeds. Seeds 123 and 456 produce identical outputs on the same GPU; all observed σ reflects cross-server floating-point divergence.

Model	Dataset	Mode	E-F1	R-F1	DR%
Qwen3-8B	SciERC	Unc	66.94 ± 0.08	43.25 ± 0.43	2.06 ± 0.08
Qwen3-8B	SciERC	Con	66.94 ± 0.08	43.22 ± 0.39	0.00 ± 0.00
Llama-3.1-8B	SciERC	Unc	67.99 ± 0.16	45.38 ± 0.05	1.04 ± 0.05
Llama-3.1-8B	SciERC	Con	67.99 ± 0.16	45.40 ± 0.05	0.03 ± 0.04

Beam search non-determinism. Greedy decoding is deterministic (§G above). Beam search ($k \geq 2$) introduces non-determinism from GPU floating-point accumulation order; we report single-run results. Based on the greedy cross-hardware variance ($\sigma < 0.5$ pp, Table 30), beam search variance is expected to be of similar magnitude.

200-document subset justification. The 70B LoRA evaluation on DocRED uses a 200-document subset (vs. the full 998-document dev set used for 8B models). To verify that this subset provides sufficient statistical power, we conduct a bootstrap analysis using the full 998-document 8B LoRA results as a reference population. Resampling 200 documents 5,000 times yields a relation F1 margin of error of ±3.3 points (95% CI) and a DR violation rate margin of error of ±0.45 percentage points—precise enough to support our conclusions. For zero-shot models, where DR violations are prevalent (15.0% unconstrained), a 200-document subset achieves 100% power to detect the constrained decoding effect (Cohen's $h = 0.80$); for fine-tuned models, where violations are already rare (<0.5%), 94.9% of 200-document subsamples correctly identify the direction of DR reduction, and the primary effect is established on the full 998-document evaluation.

Statistical power across evaluation settings. Table 31 reports bootstrap confidence interval width and minimum detectable effect (MDE) for F1 across evaluation settings (≥1,000 bootstrap resamples; MDE at 80% power, α=0.05, paired test, ρ=0.9).

WikiEvents' n=20 yields an Arg-F1 margin of error of ±3.0 pp—comparable to the 200-document DocRED subset (±3.3 pp)—with MDE≈2.0 pp.

Dataset	n	95% CI (F1)	MDE (F1)
SciERC	551	±1.8 pp	1.2 pp
DocRED	998	±0.8 pp	0.5 pp
DocRED (70B)	200	±3.3 pp	2.1 pp
WikiEvents	20	±3.0 pp	2.0 pp

Table 31: Statistical precision by dataset size. 95% CI: bootstrap confidence interval half-width for per-configuration F1. MDE: minimum detectable F1 difference at 80% power (α=0.05, paired test, conservative ρ=0.9). SciERC/DocRED: zero-shot Rel-F1; DocRED (70B): LoRA Rel-F1 (200-doc subset); WikiEvents: LoRA Arg-F1.

However, the primary WikiEvents claims concern violation elimination, not F1 improvement: RoV drops from 8.8–38.3% to exactly 0% across both models (bootstrap 95% CI $[0\%, 0\%]$ vs. $[2.4\%, 15.4\%]$ / $[25.4\%, 46.7\%]$; power≈1.0). The small test set limits detection of subtle F1 shifts, but complete elimination of schema violations is unambiguous at n=20; the primary evidence for F1 preservation rests on SciERC (n=551) and DocRED (n=998), where CIs are ≤1.8 pp.

H Effect of Document Length on Constraint Effectiveness

To examine whether OntoCoDec's constraint effectiveness varies with document complexity, we partition test documents into three length bins (tertile split by word count) and report per-bin violation counts and F1.

Table 32: Document length breakdown on SciERC (Qwen3-8B). Tertile boundaries: Short ≤ 19 words, Medium ≤ 27, Long > 27.

Bin	Mode	ET	RV	DR	CF	Total	E-F1	R-F1
Short (196)	Uncon.	1	5	57	0	63	32.4	8.9
	Const.	0	0	5	5	10	32.4	8.9
Medium (180)	Uncon.	6	9	104	0	119	27.8	6.7
	Const.	0	0	7	15	22	27.8	6.7
Long (175)	Uncon.	3	10	104	1	118	29.9	7.0
	Const.	0	0	4	20	24	30.0	7.3

Table 33: Document length breakdown on DocRED (Qwen3-8B). Tertile boundaries: Short ≤ 160 words, Medium ≤ 208, Long > 208.

Bin	Mode	ET	RV	DR	CF	Total	E-F1	R-F1
Short (336)	Uncon.	177	1238	1021	9	2445	61.1	9.3
	Const.	0	0	495	138	633	62.6	9.6
Medium (332)	Uncon.	147	1742	1047	5	2941	59.3	7.9
	Const.	0	5	544	180	729	60.7	8.5
Long (330)	Uncon.	176	2209	1162	3	3550	53.9	5.6
	Const.	0	0	731	200	931	54.8	5.8

Table 34: Violation reduction percentage by document length bin (Qwen3-8B, S1 constraints).

	SciERC		DocRED	
Bin	Viols (U→C)	Red.%	Viols (U→C)	Red.%
Short	63 → 10	84.1	2445 → 633	74.1
Medium	119 → 22	81.5	2941 → 729	75.2
Long	118 → 24	79.7	3550 → 931	73.8

The violation reduction rate is consistent across all length bins: 80–84% for SciERC and 74–75% for DocRED. Longer documents produce more violations in the unconstrained setting (e.g., DocRED long-bin: 3,550 vs. short-bin: 2,445), yet OntoCoDec achieves proportionally similar reduction; the constraint framework scales reliably with document complexity. ET and RV violations are fully eliminated across all bins. F1 scores are preserved or slightly improved under constrained decoding, with DocRED showing +0.9–1.5 E-F1 and +0.2–0.6 R-F1 improvement in the constrained setting.

I Prompt Sensitivity Analysis

To verify that OntoCoDec's constraint enforcement is robust to prompt variation, we evaluate three prompt variants on SciERC using Qwen3-8B (551 test documents): **Default** (the prompt used in all main experiments), **Minimal** (a shortened version listing only entity/relation types), and **Detailed** (an extended version with type definitions and extraction guidelines). All three variants share the same output JSON schema; only the instruction text differs. Each variant is decoded in both unconstrained and constrained (full Stage 1) modes.

Table 35: Prompt sensitivity results on SciERC (Qwen3-8B, 551 documents). ET, RV, DR denote entity-type, relation-vocabulary, and domain-range violation rates (%), respectively. ΔDR shows the relative reduction from unconstrained to constrained decoding.

Prompt	Mode	ET%	RV%	DR%	ΔDR	E-F1	R-F1
Default	Unconstrained	0.40	1.39	15.37	94.0%	31.01	8.23
	Constrained	0.00	0.00	0.93		30.98	8.36
Minimal	Unconstrained	0.58	1.16	16.91	94.6%	31.98	7.91
	Constrained	0.00	0.00	0.92		31.99	7.85
Detailed	Unconstrained	0.22	0.45	11.07	94.2%	33.55	10.48
	Constrained	0.00	0.00	0.64		33.56	10.48

Table 35 shows that constrained decoding eliminates entity-type and relation-vocabulary violations entirely (ET and RV drop to 0%) for all three prompt variants. Domain-range violations are reduced by 94.0%–94.6%, with a standard

deviation of only 0.3 percentage points across prompts. Meanwhile, F1 scores remain effectively unchanged: the largest shift is 0.13 percentage points (R-F1, Default prompt), so constraint enforcement does not degrade extraction quality.

Notably, prompt variation has a measurable effect on baseline (unconstrained) violation rates, with DR ranging from 11.1% (Detailed) to 16.9% (Minimal). Under constrained decoding, this range narrows to 0.64%–0.93%, a 20× reduction in cross-prompt variance. This convergence is expected: because OntoCoDec enforces constraints at the token level during decoding, it operates independently of prompt content. The consistent reduction rates across diverse prompt styles confirm that the mechanism is orthogonal to prompting and generalizes without prompt-specific tuning. Schema-informed prompting results serve as an upper bound on zero-shot prompt-level intervention; training-based approaches (e.g., GoLLIE) achieve compliance through code-generation format but with different coverage trade-offs (§6).

J Error Analysis and Case Studies

We present four representative cases illustrating constraint behavior across violation types. Cases 1–3 demonstrate OntoCoDec's corrective mechanisms; Case 4 illustrates a limitation.

J.1 Domain-Range Correction

Input (SciERC, Qwen3-8B): *"We present a novel model-based approach to recover the shapes and the poses of transparent objects from known motion."*

The unconstrained model assigns OTHERSCIENTIFICTERM to "shapes" and "poses," producing FEATURE-OF(TASK, OTHERSCIENTIFICTERM) triples that violate domain/range constraints. The semantic layer redirects the tail type to TASK, yielding schema-compliant triples (Table 36).

J.2 Dual-Layer Vocabulary and Type Correction

Input (DocRED, Qwen3-8B): *"Skai TV is a Greek free-to-air television network based in Piraeus. It is part of the Skai Group, one of the largest media groups in the country."*

The unconstrained model generates natural-language relation names (e.g., "country of origin") instead of Wikidata P-codes, triggering RV violations. The grammar layer forces valid P-codes; the semantic layer ensures type compatibility (Table 37).

Table 36: OntoCoDec corrects three Feature-of DR violations. Red/✗ = violation; **green/bold**/✓ = corrected (shape markers provided for colorblind accessibility).

Triple	Head	Tail	DR
Unconstrained (3 violations)			
Feature-of(recover, shapes)	Task	OtherSciTerm	✗
Feature-of(recover, poses)	Task	OtherSciTerm	✗
Feature-of(recover, transp. obj.)	Task	OtherSciTerm	✗
Used-for(model approach, recover)	Method	Task	✓
OntoCoDec (0 violations)			
Feature-of(recover, shapes)	Task	Task	✓
Feature-of(recover, poses)	Task	Task	✓
Used-for(model approach, recover)	Method	Task	✓

Table 37: Layer 1 eliminates RV violations; Layer 2 corrects DR types on DocRED.

Unconstrained Output	Violation	Type
rel="country of origin"	Not in schema	RV
rel="headquarters location"	Not in schema	RV
rel="part of"	Not in schema	RV
rel="location of formation"	Not in schema	RV
OntoCoDec Output		
rel=P495 (Skai TV, Greece)	ORG→LOC	✓
rel=P159 (Skai TV, Piraeus)	ORG→LOC	✓
rel=P127 (Skai TV, Skai Group)	ORG→ORG	✓
rel=P17 (Piraeus, Greece)	LOC→LOC	✓

J.3 Scale Effect on Schema Compliance

Input (SciERC): *"This paper investigates the utility of applying standard MT evaluation methods (BLEU, NIST, WER and PER) to building classifiers to predict semantic equivalence and entailment."*

Both Qwen3-8B and Llama-3.1-70B-Instruct (AWQ INT4) mistype "MT evaluation methods" as METHOD (gold: METRIC), but the downstream constraint behavior differs (Table 38). The 70B model produces fewer violations (2 vs. 4) and chooses a different relation (Part-of vs. Hyponym-of), yet OntoCoDec corrects both.

J.4 Limitation: Trajectory Divergence under Constraints

Input (SciERC, Mistral-7B): *"Amorph recognizes NE items in two stages: dictionary lookup*

Table 38: 8B vs. 70B on the same input. OntoCoDec corrects violations at both scales; CF arises when type corrections conflict with entity-list assignments.

Setting	Relation	Types	DR
Qwen3-8B Unconstrained			
Hyponym-of(MT eval, BLEU)	Meth→Metric	✗	
Hyponym-of(MT eval, NIST)	Meth→Metric	✗	
Hyponym-of(MT eval, WER)	Meth→Metric	✗	
Hyponym-of(MT eval, PER)	Meth→Metric	✗	
Qwen3-8B OntoCoDec			
Hyponym-of(MT eval, BLEU)	Meth→Meth	✓	
Hyponym-of(MT eval, NIST)	Meth→Meth	✓	
CF: BLEU typed Metric in entity list, Method in relation			
Llama-3.1-70B Unconstrained			
Part-of(BLEU, MT eval)	Metric→Meth	✗	
Part-of(NIST, MT eval)	Metric→Meth	✗	
Llama-3.1-70B OntoCoDec			
Part-of(BLEU, MT eval)	Metric→Generic	✓	
Part-of(NIST, MT eval)	Metric→Generic	✓	

and rule application."

The gold annotation types "Amorph" as METHOD and "NE items" as OTHERSCIENTIFICTERM, with Used-for(Amorph, NE items). The unconstrained model mistypes "Amorph" but coincidentally produces the correct relation. Under OntoCoDec, modified token probabilities alter the generation trajectory entirely and produce a different (non-gold) relation (Table 39). This illustrates a fundamental property of autoregressive generation under modified distributions: constraint enforcement can change the entire output path, not just individual token choices.

Table 39: Constrained decoding changes the generation trajectory. The unconstrained model coincidentally produced the gold triple despite type errors.

Setting	Output	Gold?
Gold	Used-for(Amorph^{Meth}, NE items^{OtherSci})	–
Unconstr.	Used-for(Amorph^{OtherSci}, NE items^{Task})	✓
OntoCoDec	Used-for(Amorph^{OtherSci}, recognizes^{Gen})	✗
	Feature-of(NE items^{Task}, in two stages^{Gen})	✗

J.5 Error Analysis: Constraint Impact on Prediction Quality

We analyze errors from Qwen3-8B LoRA + S1 on SciERC (551 documents; R-F1 43.77, 460 TP, 668 FP, 514 FN) across three categories.

False negatives: constraint-blocked vs. model-limited. Of 514 false negatives, only one (0.2%) is attributable to constraint enforcement:

fn_001 (constraint-blocked): "*Experiments on the TREC Blog track test set show that both groups of credibility indicators significantly improve retrieval effectiveness.*"

Gold: Evaluate-for(retrieval effectiveness^{Metric}, credibility indicators^{OtherSci}). The unconstrained model generates this triple correctly, but under S1, conflicting type assignments to 'credibility indicators' (OtherScientificTerm in entity list vs. Method in Evaluate-for's tail) trigger a CF violation, redirecting generation to a reversed Used-for triple.

fn_004 (relation type confusion): "*Amorph recognizes NE items in two stages: dictionary lookup and rule application.*"

Gold: Part-of(dictionary lookup^{Meth}, Amorph^{Meth}). The model predicts Used-for instead of Part-of—both are schema-valid for Method→Method pairs, so constraints provide no corrective signal. This Part-of↔Used-for confusion is the dominant FN pattern.

False positives: schema-valid but incorrect. *fp_001* (wrong relation type): Same sentence as fn_004. The model generates Used-for(dictionary lookup, Amorph) where gold is Part-of. Used-for is the majority relation in SciERC (56% of all relations), causing systematic over-prediction. Since both relation types are schema-valid for the entity type pair, constraints cannot distinguish them.

fp_007 (hallucinated relation): "*The classifiers show little gain from information about meeting context.*"

No gold relation exists. The model generates Used-for(meeting context^{OtherSci}, classifiers^{Meth}), ignoring the negation in "show little gain from." The instrumental preposition "from" triggers relation generation despite the negative result.

CF violations: entity type polysemy. *cf_001* (Generic↔Method): "*A basic reason behind this is that the overall approach extends the approach described in a companion paper...*"

The entity 'approach' receives Generic in one Used-for relation and Method in another. The gold type is Generic; the polysemy of 'approach' (generic reference vs. specific method) causes per-relation type assignment to diverge.

cf_005 (Material↔Task): "*Experimental evaluations on two challenging person re-identification datasets demonstrate the superiority of the proposed model...*"

The compound noun 'person re-identification datasets' contains both a task component and a data component. The model assigns Material in one Evaluate-for relation and Task in another within the same sentence.

Summary. Structural constraints eliminate type-invalid outputs (DR→0.09%), but semantic ambiguity within schema-valid space—relation type confusion, entity type polysemy—remains the dominant error source. Only 1 of 514 false negatives (0.2%) was attributable to constraint enforcement, confirming that the recall cost of type constraints is negligible.

Downstream impact of violation removal. A natural concern with schema-constrained decoding is whether enforcing ontological validity removes *lucky violations*—predictions that happen to be correct (true positives) yet violate the schema. To quantify this trade-off, we audit all true-positive relation triples produced by unconstrained fine-tuned models and check whether any of them carry a domain–range (DR) violation, i.e., an entity-type pair $\langle h_\tau, t_\tau \rangle$ not sanctioned by the ontology for that relation.

Dataset	Model	TP	Lucky	% TP	R drop
SciERC	Qwen-LoRA	461	4	0.87%	0.41 pp
SciERC	Llama-LoRA	467	5	1.07%	0.51 pp
DocRED	Qwen-LoRA	5746	1	0.02%	0.01 pp
DocRED	Llama-LoRA	6226	1	0.02%	0.01 pp

Table 40: Lucky violations: true-positive triples that carry a domain–range violation. **Lucky** counts such triples; **R drop** is the recall decrease (in percentage points) if they were removed.

Across all four configurations, lucky violations account for at most 1.07% of true positives (5 out of 467 on SciERC with Llama-LoRA) and as few as 0.02% on DocRED. The corresponding recall drop is negligible—at most 0.51 percentage points—while the vast majority of violated triples that constraints remove are false positives. For example, on DocRED with Qwen-LoRA, constraints eliminate 164 schema-violating predictions but only 1 of them is actually correct; the other 163 are errors that would pollute a downstream knowledge graph.

Inspecting the lucky violations qualitatively, they arise from borderline entity-type assignments rather than genuine schema gaps. On SciERC, the recurring pattern involves METRIC-typed entities (e.g., BLEU, NIST) appearing as heads of `Hyponym-of` relations, a pair absent from the training ontology because these metrics are rarely annotated as hyponyms. On DocRED, the single lucky violation stems from an entity typed as MISC instead of ORG ("Assassin's Creed Unity"), making its `developer` relation invalid under the schema despite being factually correct.

These findings confirm that schema-constrained decoding introduces a minimal recall cost while providing a substantial precision gain. Retaining these few lucky violations would allow schema-invalid triples into the knowledge graph, undermining the structural guarantees that motivate constrained decoding in the first place. The net effect is overwhelmingly positive: constraints act as a high-precision filter that removes far more erroneous triples than correct ones.

K Ontology Scalability

Scale	Types	Rels	Compile (s)	Overhead
Small (SciERC)	6	7	0.33	1.02×
Medium (DocRED)	6	96	0.32	1.03×
Large (synth.)	100	500	0.35	1.03×
XLarge (synth.)	200	1,000	0.42	1.03×

Table 41: Compilation time (one-time) and per-token inference overhead across ontology scales. Small and Medium use real ontologies; Large and XLarge use synthetic ontologies with random domain/range constraints. Overhead measured on Qwen3-8B, 10 documents per scale.

Full constraint compilation (schema parsing + grammar build) scales sub-linearly (0.32 s to 0.42 s for 7→1,000 relations; Table 41); within this, the XGrammar automaton accounts for ∼0.044 s and semantic constraint compilation grows linearly but remains negligible (0.4 ms to 17 ms). Inference overhead is constant at 2–3% because XGrammar uses precompiled bitmasks ($O(1)$ per token) and the semantic layer checks only the active relation's domain/range set rather than the full ontology. The practical limit is first-token uniqueness: when entity types exceed ∼200, prefix collisions in the tokenizer vocabulary may require fallback to multi-token disambiguation (§3.3).

L Full Fine-Tuning Baseline

To verify that constraint effectiveness generalizes beyond LoRA, we evaluate Qwen3-8B with full-parameter fine-tuning on SciERC (same data, 3 epochs, lr=2×10^{-4}, seed 42).

Full fine-tuning retains 2.74% DR violations (30 triples), concentrated in Part-of (14.6%), Hyponym-of (9.6%), and Evaluate-for (8.4%). S1 eliminates all 30 violations with no F1 cost, which confirms that constraint enforcement is effective regardless of the fine-tuning method. The CF↔DR trade-

	E-F1	R-F1	DR%	CF%
Full FT unconstrained	64.78	37.89	2.74	0
Full FT + S1	64.78	37.89	**0**	1.31
LoRA unconstrained	66.86	43.65	0.31	0
LoRA + S1	67.05	43.77	**0**	1.18

Table 42: Full fine-tuning vs. LoRA on SciERC (Qwen3-8B, 551 docs). Constraints eliminate DR under both methods. Full FT underperforms LoRA, likely due to an untuned learning rate (2×10^{-4}, standard for LoRA but high for full FT). F1 is unaffected by constraints in both cases.

off persists (CF 0%→1.31%), consistent with the LoRA pattern (1.18%).

M Post-hoc Type Consistency Resolution

S1 enforces domain/range constraints per-triple but does not enforce global entity type consensus, leaving residual CF violations where the same entity receives different types across triples. We evaluate a post-hoc resolution step (TCR): for each entity with conflicting type assignments, TCR selects the type that maximizes satisfaction count across all relations referencing that entity, subject to domain/range constraints. Unlike naive majority voting (which ignores relational constraints and increases DR), TCR is constraint-aware: candidate types are scored by how many of the entity's relational contexts they satisfy. TCR runs after S1 decoding completes and has access to the full document context.

		CF (%)		DR (%)		ΔE-F1	Fund.
Config	**Model**	Pre	Post	Pre	Post	(pp)	(%)
Zero-shot	Qwen	9.75	**0**	0.93	1.21	−1.29	2.04
	Llama	6.40	**0**	0.08	0.30	−1.78	1.85
	Mistral	0.67	**0**	13.36	12.87	+0.05	0
LoRA	Qwen	1.19	**0**	0	0	−0.75	0
	Llama	0.46	**0**	0.09	0.09	−0.18	0

Table 43: TCR on SciERC ($\bar{\kappa}$=0.333, 551 docs). Pre/Post: before/after TCR. ΔE-F1: entity F1 change (F1×100). Fund.: percentage of conflicting entities where no single type satisfies all relational constraints (fundamental conflicts). TCR eliminates all CF with DR increases ≤0.28 pp; fundamental conflicts are 0–2%.

Analysis. TCR effectiveness varies inversely with schema constraint tightness $\bar{\kappa}$ (descriptive pattern across two schemas). On SciERC ($\bar{\kappa}$=0.333), fundamental conflicts constitute 0–2% of type disagreements: most conflicts are resolvable because the loose schema admits many valid type pairs,

		CF (%)		DR (%)		ΔE-F1	Fund.
Config	**Model**	Pre	Post	Pre	Post	(pp)	(%)
Zero-shot	Qwen	13.11	**0**	10.21	11.74	−3.44	13.52
	Llama	23.71	**0**	2.55	11.93	−5.21	22.78
	70B	11.61	**0**	0	4.20	−5.98	18.26
LoRA	Qwen	0.42	**0**	0.01	0.14	−0.16	12.16
	Llama	0.36	**0**	0.21	0.20	−0.17	11.59

Table 44: TCR on DocRED ($\bar{\kappa}$=0.842, 998 docs). Under zero-shot, TCR introduces substantial DR regressions (up to +9.38 pp for Llama) and E-F1 drops (up to −5.98 pp for 70B). Fundamental conflict rates are 12–23%, an order of magnitude higher than SciERC. Under LoRA, low baseline CF (0.36–0.42%) limits both TCR impact and DR cost.

and TCR can find a type satisfying all relational contexts. DR increases are bounded (≤0.28 pp), and E-F1 cost is modest (≤1.78 pp in zero-shot, ≤0.75 pp under LoRA). On DocRED ($\bar{\kappa}$=0.842), 13–23% of conflicts are fundamental: the tight schema renders most type pairs invalid, so unifying an entity's type across all its relational contexts inevitably violates some domain/range constraints. This produces severe DR regressions under zero-shot (Llama: 2.55%→11.93%; 70B: 0%→4.20%), where baseline CF is high enough that many entities require resolution. Under LoRA, baseline CF is already low (0.36–0.42%), limiting both the scope for TCR improvement and the DR cost.

This $\bar{\kappa}$-dependent boundary is consistent with the constraint tightness analysis in §4: $\bar{\kappa}$ characterizes not only the decoding-time DR residual regime but also the feasibility of post-hoc type resolution. Beam search (§5.4) resolves CF and DR simultaneously across all $\bar{\kappa}$ values (SciERC and DocRED (998 docs): DR→0% for calibrated models), making it the preferred approach for high-tightness schemas. TCR is a zero-cost complement for low-tightness schemas where beam search latency (1.54×) is undesirable.

N Constraint Tightness Case Study

Analysis. The case study confirms that violation susceptibility is driven by the interaction of constraint tightness and entity type confusability, not tightness alone. Among DocRED relations with $\kappa \geq 0.81$, those involving semantically unambiguous type pairs (PER→LOC for kinship/geographic relations) achieve DR=0%, while those requiring the model to distinguish frequently confused types (LOC vs. ORG, PER vs. MISC) show DR>50%. On SciERC, the highest-DR relation (COMPARE,

Relation	Dataset	κ	DR%	n	Explanation
Highest DR violation rate					
basin country	DocRED	.97	74.6	63	Sole valid pair LOC→LOC; head mistyped as ORG/MISC
product produced	DocRED	.94	67.2	61	Requires ORG head; model assigns LOC/PER
participant of	DocRED	.81	58.6	326	Hallucinated EVENT type; PER↔event confusion
COMPARE	SciERC	.53	42.4	66	Cross-category pairs violate same-type constraint
FEATURE-OF	SciERC	.28	33.5	486	Fuzzy Task/OtherScientificTerm boundary
High κ with zero violations (DR = 0%)					
lyrics by	DocRED	.94	0.0	9	Unambiguous PER→MISC; creative roles rarely mistyped
place of birth	DocRED	.92	0.0	157	Unambiguous PER→LOC; no type confusion
place of death	DocRED	.89	0.0	60	Strong PER→LOC prior; mirrors place of birth
subsidiary	DocRED	.86	0.0	5	Clear ORG→ORG corporate semantics
father	DocRED	.81	0.0	69	Clear PER→PER kinship semantics

Table 45: Per-relation κ vs. DR violation rate (Qwen3-8B, zero-shot, unconstrained). High κ is necessary but not sufficient for high DR: relations with unambiguous entity types (place of birth, κ=0.92) have zero violations, while equally tight relations with confusable types (basin country, κ=0.97) show 74.6% DR.

42.4%) has only moderate κ (0.53) but high type confusability across scientific categories. This two-factor pattern motivates constrained decoding: the largest gains come from high-κ, confusable-type relations where the model's type uncertainty is channeled into valid pairs.

Schema ambiguity analysis. We define the *schema ambiguity* of a relation r as the average fraction of relations that share each of r's valid type pairs:

$$\alpha_r = \frac{1}{|\mathcal{P}_r|} \sum_{(h,t)\in\mathcal{P}_r} \frac{|\{r' \in \mathcal{R} : (h,t) \in \mathcal{P}_{r'}\}|}{|\mathcal{R}|} \quad (3)$$

where $\mathcal{P}_r$ is the set of valid entity type pairs for relation r and $\mathcal{R}$ is the full relation set. High α_r indicates that r's valid type pairs are shared by many other relations, limiting the discriminative power of type constraints for that relation.

Table 46 reveals a structural asymmetry between datasets. In SciERC, all seven relations have $\alpha \geq 0.67$: with only 7 relations over $6^2 = 36$ entity type pair slots, extensive overlap is inevitable. For example, the pair (Method, Method) is valid for all 7 relations ($\alpha = 1.0$). In DocRED, 96 relations spread across the same 36 type pair slots yield much lower per-relation ambiguity; only 12 relations exceed $\alpha \geq 0.36$, and these collectively cover just 3.8% of gold triples.

Table 47 shows the DR impact stratified by ambiguity. On DocRED, high-α relations exhibit substantially higher unconstrained DR (43.8% for Llama vs. 29.9% for low-α), reflecting the inherent difficulty of relations whose valid type pairs are shared by dozens of other relations—the model frequently assigns a type combination that, while valid for *some* relation, violates the constraints of the target relation. Under S1 constraints, Llama

Table 46: Schema ambiguity α for all SciERC relations and the 12 DocRED relations with $\alpha \geq 0.36$. **Gold %**: fraction of all gold triples belonging to this relation. All SciERC relations exceed the threshold due to the small schema (7 relations, 6 entity types); in DocRED, only 12 of 96 relations qualify, covering 3.8% of gold triples.

Dataset	Relation	α	$\lvert\mathcal{P}_r\rvert$	Gold	Gold %
SciERC (all 7 relations exceed $\alpha \geq 0.36$)					
SciERC	Hyponym-of	0.83	21	298	9.3
SciERC	Part-of	0.83	21	179	5.6
SciERC	Compare	0.82	17	166	5.2
SciERC	Conjunction	0.77	25	400	12.4
SciERC	Feature-of	0.76	26	173	5.4
SciERC	Evaluate-for	0.71	23	313	9.7
SciERC	Used-for	0.67	35	1,690	52.5
DocRED (12 of 96 relations with $\alpha \geq 0.36$; 3.8% of gold)					
DocRED	sister city (P190)	0.42	1	4	0.01
DocRED	basin country (P205)	0.42	1	85	0.22
DocRED	mouth of watercourse (P403)	0.42	1	95	0.25
DocRED	separated from (P807)	0.41	2	2	0.01
DocRED	replaces (P1365)	0.39	4	18	0.05
DocRED	capital of (P1376)	0.37	3	76	0.20
DocRED	place of birth (P19)	0.37	3	511	1.34
DocRED	operator (P137)	0.36	7	95	0.25
DocRED	military branch (P241)	0.36	5	108	0.28
DocRED	production company (P272)	0.36	5	82	0.21
DocRED	place of death (P20)	0.36	4	203	0.53
DocRED	location (P276)	0.36	5	172	0.45

Table 47: DR violation rates partitioned by schema ambiguity. High-α relations ($\alpha \geq 0.36$) start with substantially higher unconstrained DR on DocRED, and retain elevated residual DR under S1 for Qwen but not for Llama.

		Unconstrained		S1 (Constrained)	
Model	**Partition**	DR%	n	DR%	n
Llama-3.1-8B	High-α (12 rels)	43.8	861	3.2	1,043
	Low-α (84 rels)	29.9	12,809	2.5	15,325
Qwen3-8B	High-α (12 rels)	31.9	718	20.0	1,119
	Low-α (84 rels)	21.2	10,898	9.5	16,217

reduces both groups to near-parity (3.2% vs. 2.5%), whereas Qwen retains a 2× gap (20.0% vs. 9.5%), indicating that constraint effectiveness on high-ambiguity relations depends on the model's baseline type accuracy. Excluding the 12 high-α relations from the DocRED evaluation changes the overall S1 DR only marginally—from 2.55% to 2.51% for Llama and from 10.21% to 9.53% for Qwen—confirming that their small gold footprint (3.8%) limits their aggregate impact despite elevated per-relation rates. On SciERC, where all relations have high α, the exclusion analysis is vacuous; yet OntoCoDec still reduces DR from 15.4% to 0.9% (Qwen) and from 13.3% to 0.08% (Llama), demonstrating that high schema ambiguity does not preclude effective constraint enforcement.

Order invariance. To verify that OntoCoDec's outputs are independent of schema enumeration

order, we randomly shuffle the entity and relation type lists in the compiled constraint file (`ontology_constraints.json`) and re-run greedy S1 decoding on all 551 SciERC documents. The raw model outputs and parsed triples are byte-identical to the canonical-order run across all documents. This holds because the grammar layer accepts the same token set at each type-bearing position regardless of the order in which types are listed in the schema, and greedy decoding is deterministic given identical logit masks. The result confirms that OntoCoDec introduces no spurious order dependence from schema compilation.

O Full Proof of Proposition 1

We prove that domain/range type constraints for IE with unbounded entity sets cannot be expressed by any context-free grammar, using the pumping lemma for context-free languages.

Proof. Define the language L of valid IE outputs over ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$ with $\mathcal{E} = \{\tau_A, \tau_B\}$, $\mathcal{R} = \{r\}$, and $\sigma(r) = \{(\tau_A, \tau_A)\}$ (i.e., r requires both head and tail to be type τ_A).

An output string has the form:

$$w = e_1\, e_2 \cdots e_n \;\; \texttt{REL}(h{=}e_k,\, r,\, \tau^h)$$

where each e_i is an entity with type $\tau_{a_i} \in \{\tau_A, \tau_B\}$, and τ^h is the head type field of a relation triple referencing entity e_k ($1 \le k \le n$).

The type-validity constraint requires $\tau^h = \tau_{a_k}$ (the head type must match the referenced entity's type) *and* $\tau^h \in \mathrm{dom}(r) = \{\tau_A\}$. Combined: a string $w \in L$ if and only if $\tau_{a_k} = \tau_A$.

Now consider strings of the specific form:

$$w_n = \underbrace{\tau_B\; \tau_B \;\cdots\; \tau_B}_{n-1}\; \tau_A \;\; \texttt{REL}(h{=}e_n,\, r,\, \tau_A)$$

where the first $n{-}1$ entities have type τ_B and only entity e_n has type τ_A. The relation references e_n, so $\tau^h = \tau_A = \tau_{a_n}$, satisfying both the registry consistency and domain constraint. Thus $w_n \in L$ for all n.

Assume for contradiction that L is context-free. By the pumping lemma, there exists a pumping length p such that any $w \in L$ with $|w| \ge p$ can be written as $w = uvxyz$ where:

(i) $|vy| \ge 1$,

(ii) $|vxy| \le p$,

(iii) $uv^ixy^iz \in L$ for all $i \ge 0$.

Choose $w = w_n$ with $n > p$. The entity list portion $\tau_B^{n-1}\,\tau_A$ has length $n > p$, so the pumpable region vxy (with $|vxy| \le p$) must lie entirely within the entity list (it cannot span from the entity list to the relation reference, which is separated by $n - p > 0$ entity tokens from the start).

Case 1: v or y contains entity tokens from the τ_B^{n-1} prefix. Pumping up ($i = 2$): uv^2xy^2z inserts additional τ_B entities, increasing n to some $n' > n$. But the relation reference still points to entity e_n (at the original position), which is now at a different position in the longer entity list. The entity at position n in the pumped string is τ_B (since we inserted τ_B tokens before position n, shifting τ_A to position $n' > n$). Thus $\tau^h = \tau_A \neq \tau_B = \tau_{a'_n}$, where n' is the new position of entity e_n. More precisely: the relation references a fixed entity name e_k, but pumping changes the entity-type binding at that name's position. Since the pumped string assigns τ_B at position k (which originally had τ_A), the type-validity constraint $\tau^h = \tau_{a_k} \in \mathrm{dom}(r)$ is violated. Hence $uv^2xy^2z \notin L$, contradicting condition (iii).

Case 2: v or y contains the τ_A token at position n. Pumping down ($i = 0$): uxz removes the only τ_A entity. The relation's head entity reference now points to a position that either does not exist or has type τ_B. In either case, $\tau^h = \tau_A \neq \tau_B$, violating type validity. Hence $uxz \notin L$, contradicting condition (iii).

In both cases, the pumping lemma is violated, so L is not context-free. The key insight is that type-validity requires a *binding* between an entity's type (assigned at an earlier position) and the relation's type field (at a later position), with the binding target determined by the entity name. Pumping disrupts this binding by changing which entity occupies a given position, while the relation reference remains fixed—a cross-position dependency that pushdown automata cannot track with their LIFO stack discipline.

OntoCoDec's semantic layer addresses this with a runtime entity registry $\mathcal{D}$ (a finite map), operating as a deterministic stateful transducer that provides random-access type lookups, beyond the expressive power of context-free grammars. □

Finite entity sets. For a finite entity set $\mathcal{E}$ with canonical identifiers, the valid language $L_{\text{valid}} = \bigcap_i L_i$ is theoretically regular (a finite intersection of regular languages). However, the corresponding

DFA requires $O(|\mathcal{T}|^{|\mathcal{E}|})$ states—for SciERC with 6 entity types and typical documents containing 20+ entities, this exceeds 10^{15} states, rendering explicit construction impractical. Moreover, real IE operates over open-vocabulary text spans rather than closed identifier sets, making the general (non-regular) formulation the appropriate model. Proposition 1 motivates the runtime state-tracking architecture for this general case.

Position indexing vs. name referencing. The proof above uses position-based entity indexing (e_k at position k) for expository clarity. In OntoCoDec's actual JSON output, relation triples reference entities by name strings rather than positional indices. The core argument transfers directly: validating type consistency requires a name→type lookup over dynamically generated bindings, which is a random-access map operation. A PDA's LIFO stack cannot perform random-access retrieval by key (it can only inspect the top element), so the non-context-free result holds regardless of whether entities are referenced by position or by name. Name-based referencing strictly subsumes position-based indexing: any position-based constraint can be expressed via names (assign names $e_1, \ldots, e_n$ in order), but name-based constraints additionally permit cross-field coreference (the same entity name appearing in multiple triples must carry the same type), which position indexing cannot express. Thus, the position-based formulation is a simplification that makes the pumping argument more transparent without weakening the conclusion.